Supplementary Materials: Input uncertainty propagation through trained neural networks

1. Introduction

This document is the supplementary materials file of the main paper ”Input uncertainty propagation through trained neural networks”. It contains the derivation proofs for analytical formula of first two moments of a multivariate Gaussian distribution through ReLU, Leaky Relu and ELU activation functions.

2. Analytical moment propagation through classical activation functions.

For fully connected network and CNN, we provide the analytical formula to propagate the first two moments of a Gaussian distribution through ReLU (Fukushima, 1975), leaky ReLU (Maas et al., 2013) or ELU (Clevert et al., 2015) activation function.
Due to the excessive length of the proofs, they are detailled on the supplementary files available at https://github.com/PaulMcht/WGMprop.

2.1. ReLU activation function

Proposition 2.1. *Let X be a random variable with $X \sim \mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\mu} \in \mathbb{R}^d$, $\boldsymbol{\Sigma} \in \mathbb{S}_d^+(\mathbb{R})$ with rank$(\boldsymbol{\Sigma}) = r$, $r \in \mathbb{N}^*$ and f be the rectified linear function, then for the random variable $\boldsymbol{Y} = (Y_1, ..., Y_d) = f(\boldsymbol{X}) = (f(X_1), ..., f(X_d))$, $\forall k, k' \in \{1...d\}$:*

$$\begin{aligned}
\mathbb{E}[Y_k] &= c_k\phi_k + \mu_k\Phi_k && (1)\\
\mathbb{E}[Y_k^2] &= (\mu_k^2 + c_k^2)\Phi_k + \mu_k c_k \phi_k && (2)\\
\mathbb{E}[Y_k Y_{k'}] &= c_k\alpha\phi_{k'}\phi_{NS} + (\mu_k\mu_{k'} + c_k\beta)\left(\Phi_{k'} - \Phi_{kk'}\right) + \mu_k c_{k'}\phi_{k'}\Phi_S + \mu_{k'}c_k\phi_k\Phi_{NS} && (3)
\end{aligned}$$

where: $\mu_k = \mathbb{E}[X_k]$, the mean of the kth component of the random variable $\boldsymbol{X}$, $Q_r = (q_k)_{k\in\{1...d\}} \in \mathcal{M}_{r,d}(\mathbb{R})$ the r first line of Q such that $\Sigma = Q^T\Lambda Q = Q_r^T\Lambda_r Q_r$, and $\Lambda_r \in \mathcal{D}_r(\mathbb{R})$. Then, we have $c_k = \left\|\sqrt{\Lambda_r}q_k\right\|$, $c_{k-k'} = \left\|\sqrt{\Lambda_r}(q_k - q_{k'})\right\|$, $\beta = \frac{c_k^2+c_j^2-c_{k-k'}^2}{2c_k}$, $\alpha = \sqrt{c_j^2 - \beta^2}$. Finally, we note $\phi_k = \phi_{0,1}\left(\frac{\mu_k}{c_k}\right)$, $\Phi_{k'} = \Phi_{0,1}\left(\frac{\mu_{k'}}{c_{k'}}\right)$, $\Phi_{NS} = \Phi_{0,1}\left(\frac{\mu_{k'}}{\alpha} - \frac{\beta\mu_k}{\alpha c_k}\right)$, $\Phi_S = \Phi_{0,1}\left(\frac{c_{k'}\mu_k}{c_k\alpha} - \frac{\beta\mu_{k'}}{\alpha c_{k'}}\right)$, $\phi_{NS} = \phi_{0,1}\left(\frac{\beta\mu_{k'}}{\alpha c_{k'}} - \frac{\mu_k c_{k'}}{c_k\alpha}\right)$ and $\Phi_{kk'} = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_{k'}}\\ -\frac{\beta}{c_{k'}} & 1\end{bmatrix}}\left(\left[\frac{\mu_{k'}}{c_{k'}}, -\frac{\mu_k}{c_k}\right]^T\right)$ and ϕ and Φ are respectively the PDF and CDF of the standard normal distribution.

2.2. Leaky ReLU activation function

Proposition 2.2. *Let X be a random variable with $X \sim \mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\mu} \in \mathbb{R}^d$, $\boldsymbol{\Sigma} \in \mathbb{S}_d^+(\mathbb{R})$ with rank$(\boldsymbol{\Sigma}) = r$, $r \in \mathbb{N}^*$ and f be the leaky rectified linear function of parameters $\lambda \in [0, 1[$, then for the random variable $\boldsymbol{Y} = (Y_1, ..., Y_d) = f(\boldsymbol{X}) = (f(X_1), ..., f(X_d))$, $\forall k, k' \in \{1...d\}$:*

$$\begin{aligned}
\mathbb{E}[Y_k] &= (1-\lambda)c_k\phi_k + (1-\lambda)\mu_k\Phi_k + \lambda\mu_k && (4)\\
\mathbb{E}[Y_k^2] &= (1-\lambda^2)(\mu_k^2 + c_k^2)\Phi_k + (1-\lambda^2)\mu_k c_k\phi_k + \lambda^2(c_k^2 + \mu_k^2) && (5)\\
\mathbb{E}[Y_k Y_{k'}] &= (1-\lambda)^2 c_k\alpha\phi_{k'}\phi_{NS}\\
&+ (\mu_k\mu_{k'} + c_k\beta)\left((1-\lambda)\Phi_{k'} - (1-\lambda)^2\Phi_{kk'} + \lambda(1-\lambda)\Phi_k + \lambda^2\right)\\
&+ (1-\lambda)^2\mu_k c_{k'}\phi_{k'}\Phi_S\\
&+ (1-\lambda)^2\mu_{k'}c_k\phi_k\Phi_{NS}\\
&+ \lambda(1-\lambda)(\mu_{k'}c_k\phi_k + \mu_k c_{k'}\phi_{k'}) && (6)
\end{aligned}$$

where: $\mu_k = \mathbb{E}[X_k]$, the mean of the kth component of the random variable $\boldsymbol{X}$, $Q_r = (q_k)_{k\in\{1...d\}} \in \mathcal{M}_{r,d}(\mathbb{R})$ the r first line of Q such that $\Sigma = Q^T\Lambda Q = Q_r^T\Lambda_r Q_r$, and $\Lambda_r \in \mathcal{D}_r(\mathbb{R})$. Then, we have $c_k = \left\|\sqrt{\Lambda_r}q_k\right\|$, $c_{k-k'} = \left\|\sqrt{\Lambda_r}(q_k - q_{k'})\right\|$, $\beta = \frac{c_k^2+c_j^2-c_{k-k'}^2}{2c_k}$, $\alpha = \sqrt{c_j^2 - \beta^2}$. Finally, we note $\phi_k = \phi_{0,1}\left(\frac{\mu_k}{c_k}\right)$, $\Phi_{k'} = \Phi_{0,1}\left(\frac{\mu_{k'}}{c_{k'}}\right)$, $\Phi_{NS} = \Phi_{0,1}\left(\frac{\mu_{k'}}{\alpha} - \frac{\beta\mu_k}{\alpha c_k}\right)$, $\Phi_S = \Phi_{0,1}\left(\frac{c_{k'}\mu_k}{c_k\alpha} - \frac{\beta\mu_{k'}}{\alpha c_{k'}}\right)$, $\phi_{NS} = \phi_{0,1}\left(\frac{\beta\mu_{k'}}{\alpha c_{k'}} - \frac{\mu_k c_{k'}}{c_k\alpha}\right)$ and $\Phi_{kk'} = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_{k'}}\\ -\frac{\beta}{c_{k'}} & 1\end{bmatrix}}\left(\left[\frac{\mu_{k'}}{c_{k'}}, -\frac{\mu_k}{c_k}\right]^T\right)$ and ϕ and Φ are respectively the PDF and CDF of the standard normal distribution.

2.2.1. ELU ACTIVATION FUNCTION

Proposition 2.3. *Let X be a random variable with $X \sim \mathcal{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\boldsymbol{\mu} \in \mathbb{R}^d$, $\boldsymbol{\Sigma} \in \mathbb{S}_d^+(\mathbb{R})$ with rank$(\boldsymbol{\Sigma}) = r$, $r \in \mathbb{N}^*$ and f be the exponential linear function, then for the random variable $\boldsymbol{Y} = (Y_1, ..., Y_d) = f(\boldsymbol{X}) = (f(X_1), ..., f(X_d))$, $\forall k, k' \in \{1...d\}$:*

$$\begin{aligned}
\mathbb{E}[Y_k] &= c_k\phi_k + (\mu_k + \lambda)\Phi_k + \lambda e^{\mu_k + \frac{c_k^2}{2}}(1 - \Phi_{kc_k}) - \lambda && (7)\\
\mathbb{E}[Y_k^2] &= (\mu_i^2 + c_i^2 - \lambda^2)\Phi_i + \mu_i c_i \phi_i + \lambda^2 e^{2(c_i^2+\mu_i)}(1 - \Phi_{2c_i}) - 2\lambda^2 e^{\mu_i + \frac{c_i^2}{2}}(1 - \Phi_{c_i}) \\
&+ \lambda^2 && (8)\\
\mathbb{E}[Y_k Y_{k'}] &= c_i\alpha\phi_j\phi_{NS} \\
&+ (\mu_i\mu_j + c_i\beta + \lambda\mu_i)\,\Phi_j \\
&- \left(\mu_i\mu_j + c_i\beta + \lambda(\mu_i + \mu_j) + \lambda^2\right)\Phi_{ij} \\
&+ (\mu_i c_j\phi_j + \lambda c_j\phi_j + \frac{\lambda\beta c_i}{c_j}((\phi_j - e^{\mu_j + \frac{c_j^2}{2}}\phi_{c_j}))\Phi_S \\
&+ (\mu_j c_i\phi_i + \lambda c_i\phi_i + \lambda\beta(\phi_i - e^{\mu_i + \frac{c_i^2}{2}}\phi_{c_i}))\Phi_{NS} \\
&+ \lambda c_i e^{\mu_j + \frac{c_j^2}{2}}\phi_{\beta_i}\Phi_\alpha \\
&+ \lambda c_j e^{\mu_i + \frac{c_i^2}{2}}\phi_a\Phi_e \\
&+ \lambda^2 e^{\mu_i + \frac{c_i^2}{2}}\Phi_{c_i} \\
&- \lambda e^{\mu_j + \frac{c_j^2}{2}}(\mu_i + \beta c_i)\Phi_{c_j} \\
&+ \lambda e^{\mu_i + \frac{c_i^2}{2}}(\lambda + \beta c_i + \mu_j)\Phi_A \\
&+ \lambda e^{\mu_j + \frac{c_j^2}{2}}(\lambda + \beta c_i + \mu_i)(\Phi_{\beta_i} + \Phi_B) \\
&+ \lambda^2 e^{\mu_i + \mu_j + \frac{c_i^2 + c_j^2}{2} + c_i\beta}(1 - \Phi_{\beta + c_i} - \Phi_C) \\
&- \lambda(\lambda e^{\mu_i + \frac{c_i^2}{2}} + \lambda e^{\mu_j + \frac{c_j^2}{2}} + c_i\phi_i + c_j\phi_j - \lambda) && (9)
\end{aligned}$$

where: $\mu_k = \mathbb{E}[X_k]$, the mean of the kth component of the random variable $\boldsymbol{X}$, $Q_r = (q_k)_{k\in\{1...d\}} \in \mathcal{M}_{r,d}(\mathbb{R})$ the r first line of Q such that $\Sigma = Q^T\Lambda Q = Q_r^T\Lambda_r Q_r$, and $\Lambda_r \in \mathcal{D}_r(\mathbb{R})$. Then, we have $c_k = \left\|\sqrt{\Lambda_r}q_k\right\|$, $c_{k-k'} = \left\|\sqrt{\Lambda_r}(q_k - q_{k'})\right\|$, $\beta = \frac{c_k^2 + c_j^2 - c_{k-k'}^2}{2c_k}$, $\alpha = \sqrt{c_j^2 - \beta^2}$. Finally, we note $\phi_k = \phi_{0,1}\left(\frac{\mu_k}{c_k}\right)$, $\Phi_{k'} = \Phi_{0,1}\left(\frac{\mu_{k'}}{c_{k'}}\right)$, $\Phi_{NS} = \Phi_{0,1}\left(\frac{\mu_{k'}}{\alpha} - \frac{\beta\mu_k}{\alpha c_k}\right)$, $\Phi_S = \Phi_{0,1}\left(\frac{c_{k'}\mu_k}{c_k\alpha} - \frac{\beta\mu_{k'}}{\alpha c_{k'}}\right)$, $\phi_{NS} = \phi_{0,1}\left(\frac{\beta\mu_{k'}}{\alpha c_{k'}} - \frac{\mu_k c_{k'}}{c_k\alpha}\right)$, $\Phi_{kk'} = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_{k'}}\\ -\frac{\beta}{c_{k'}} & 1\end{bmatrix}}\left(\left[\frac{\mu_{k'}}{c_{k'}}, -\frac{\mu_k}{c_k}\right]^T\right)$, $\phi_a = \phi_{0,1}\left(\frac{\mu_j + \beta c_i}{c_j}\right)$, $\Phi_{\beta_i} = \Phi_{0,1}\left(\frac{\mu_i}{c_i} + \beta\right)$, $\Phi_e = \Phi_{0,1}\left(\frac{\beta}{\alpha}\frac{\mu_j + \beta c_i}{c_j} - \frac{c_j\mu_i}{\alpha c_i} - \frac{c_j c_i}{\alpha}\right)$, $\Phi_\alpha = \Phi_{0,1}\left(\frac{\beta\mu_i}{\alpha c_i} - \frac{\mu_j}{\alpha} - \alpha\right)$, $\Phi_{\beta + c_i} = \Phi_{0,1}(\frac{\mu_i}{c_i} + c_i + \beta)$, $\Phi_{c_i} = \Phi_{0,1}\left(\frac{\mu_i}{c_i} + c_i\right)$, $\Phi_{2c_i} = \Phi_{0,1}\left(\frac{\mu_i}{c_i} + 2c_i\right)$, $\Phi_A = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j + \beta c_i}{c_j}, -\frac{\mu_i}{c_i} - c_i\right]^T\right)$,

$\Phi_B = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j} + c_j, -\frac{\mu_i}{c_i} - \beta\right]^T\right)$,

and $\Phi_C = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j + c_j^2 + \beta c_i}{c_j}, -(\frac{\mu_i}{c_i} + c_i + \beta)\right]^T\right)$),

Finally, ϕ and Φ are respectively the PDF and CDF of the standard normal distribution.

3. Derivation for Relu

3.1. Expected value estimation

We wish to estimate the moments of the distribution when passed through an elementary block composed of a linear operation and a ReLU function. Let's note $W \in \mathbb{R}^{n\times d}$ and $B \in \mathbb{R}^d$ the parameters of the linear operation such that $X_l = WX + B \in \mathbb{R}^n$ and the ReLU function such that: $Y = \mathrm{ReLU}(X_l) \in \mathbb{R}^n$

The linear operation transforms a multivariate Gaussian to an other multivariate Gaussian distribution, so:

$$X_l \sim \mathcal{N}(\mu_l, \Sigma_l),$$

$$\begin{cases} \mu_l = W\mu + B, \\ \Sigma_l = W\Sigma W^T, \end{cases}$$

with $\mathrm{rank}(\Sigma_l) \leq \min(d, \mathrm{rank}(W))$. $\Sigma_l \in \mathcal{S}_n^+(\mathbb{R})$, so by diagonalisation we have

$$\Sigma_l = Q\Lambda Q^T,$$

where $Q \in \mathbb{O}_n(\mathbb{R})$. Let's note r the rank of Σ_l, then, there exists a matrix Q_r such that:

$$\Sigma_l = Q_r \Lambda_r Q_r^T,$$

where $Q_r \in \mathbb{R}^{n\times r}$ is the matrix composed of the r first columns of Q and $\Lambda \in \mathbb{R}^{r\times r}$ the diagonal matrix containing the r strictly positive eigenvalues of Σ_l.
We note the random variable $R = Q_r Z + \mu_l$ where $Z \sim \mathcal{N}(0, \Lambda_r)$ ie $R \sim \mathcal{N}(\mu_l, \Sigma_l)$. X_l and R are equal in term of distribution.
Let's note f_i such that:

$$\begin{aligned} f_i \colon \mathbb{R}^n &\to \mathbb{R}_+ \\ x &\mapsto x_i \mathbb{1}_{x_i \geq 0} \end{aligned}$$

Then:

$$\begin{aligned} \mathbb{E}[Y_i] &= \mathbb{E}_{z\sim p(Z)}\left[f_i(Q_r z + \mu)\right] \\ &= \mathbb{E}_{z\sim p(Z)}\left[(z^T q_i + \mu_i)\mathbb{1}_{(z^T q_i + \mu_i \geq 0)}\right] \\ &= \int_{-\infty}^{+\infty} (z^T q_i + \mu_i)\mathbb{1}_{(z^T q_i + \mu_i \geq 0)} \phi_{0,\Lambda_r}(z) dz \end{aligned} \tag{10}$$

where ϕ_{0,Λ_r} is the probability density function of the multivariate Gaussian distribution with zero mean and covariance matrix Λ_r.

Let's note

$$Z_i^+ = \left\{ z \in \mathbb{R}^r : z^T q_i + \mu_i \geq 0 \right\},$$

then

$$\mathbb{E}[Y_i] = \int_{Z_i^+} z^T q_i \phi_{0,\Lambda_r}(z) dz + \mu_i \int_{Z_i^+} \phi_{0,\Lambda_r}(z) dz \tag{11}$$

With:

$$\int_{Z_i^+} z^T q_i \phi_{0,\Lambda_r}(z) dz = (2\pi)^{-\frac{r}{2}} |\Lambda_r|^{-\frac{1}{2}} \int_{Z_i^+} z^T q_i \exp\left(-\frac{1}{2} z^T \Lambda_r z\right) dz \tag{12}$$

We note $a = \sqrt{\Lambda_r^{-1}} z$, ie $z = \sqrt{\Lambda_r} a$; $|J| = |\Lambda_r|^{\frac{1}{2}}$ and

$$\bar{Z}_i^+ = \left\{ a \in \mathbb{R}^r : a^T \sqrt{\Lambda_r} q_i + \mu_i \geq 0 \right\}$$

We need to perform an integration over an half space of $\mathbb{R}^r$. Then:

$$\int_{Z_i^+} z^T q_i \phi_{0,\Lambda_r}(z) dz = (2\pi)^{-\frac{r}{2}} \int_{\bar{Z}_i^+} a^T \sqrt{\Lambda_r} q_i \exp\left(-\frac{1}{2} a^T a\right) da \tag{13}$$

By performing a rotation with respect to the hyperplan equation $a^T q_i = 0$, we find an unitary matrix B such that $B\sqrt{\Lambda_r} qi = \left\| \sqrt{\Lambda_r} q_i \right\| e_r$ where $e_r = [0, 0, ..., 1]^T \in \mathbb{R}^r$, then

$$\begin{aligned} \tilde{Z}_i^+ &= \left\{ b \in \mathbb{R}^r : b \left\| \sqrt{\Lambda_r} q_i \right\| e_r + \mu_i \geq 0 \right\} \\ &= \left\{ b \in \mathbb{R}^r : b_r \geq -\frac{\mu_i}{\left\| \sqrt{\Lambda_r} q_i \right\|} \right\} \end{aligned} \tag{14}$$

$$\begin{aligned} \int_{Z_i^+} z^T q_i \phi_{0,\Lambda_r}(z) dz &= (2\pi)^{-\frac{r}{2}} \int_{\tilde{Z}_i^+} b^T \left\| \sqrt{\Lambda_r} q_i \right\| e_r \exp\left(-\frac{1}{2} b^T b\right) db \\ &= (2\pi)^{-\frac{r}{2}} \int_{-\infty}^{+\infty} ... \int_{-\infty}^{+\infty} \int_{-\frac{\mu_i}{\left\| \sqrt{\Lambda_r} q_i \right\|}}^{+\infty} \left\| \sqrt{\Lambda_r} q_i \right\| b_r \exp\left(-\frac{1}{2} b^T b\right) db_r \\ &= \left\| \sqrt{\Lambda_r} q_i \right\| (2\pi)^{-\frac{1}{2}} \int_{-\frac{\mu_i}{\left\| \sqrt{\Lambda_r} q_i \right\|}}^{+\infty} b_r \exp\left(-\frac{1}{2} b_r^2\right) db_r \\ &= \left\| \sqrt{\Lambda_r} q_i \right\| \phi_{0,1}\left(\frac{\mu_i}{\left\| \sqrt{\Lambda_r} q_i \right\|}\right) \end{aligned} \tag{15}$$

Similarly:

$$\begin{aligned} \int_{Z_i^+} \phi_{0,\Lambda_r}(z) dz &= \int_{-\frac{\mu_i}{\left\| \sqrt{\Lambda_r} q_i \right\|}}^{+\infty} \phi_{0,1}(b_r) db_r \\ &= \Phi_{0,1}\left(\frac{\mu_i}{\left\| \sqrt{\Lambda_r} q_i \right\|}\right) \end{aligned} \tag{16}$$

Then, by noting $c_i = \left\| \sqrt{\Lambda_r} q_i \right\|$, we have:

$$\boxed{\mathbb{E}[Y_i] = c_i \phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \mu_i \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)} \tag{17}$$

3.2. Covariance value estimation

Let's note f_{ij} such that:

$$\begin{aligned} f_{ij} \colon \mathbb{R}^n &\to \mathbb{R}_+ \\ x &\mapsto x_i \mathbb{1}_{x_i \geq 0} x_j \mathbb{1}_{x_j \geq 0} \end{aligned}$$

$$\begin{aligned} \mathbb{E}[Y_i Y_j] &= \mathbb{E}_{z \sim p(Z)} \left[f_{ij}(Q_r z + \mu) \right] \\ &= \mathbb{E}_{z \sim p(Z)} \left[(z^T q_i + \mu_i) \mathbb{1}_{(z^T q_i + \mu_i \geq 0)} (z^T q_j + \mu_j) \mathbb{1}_{(z^T q_j + \mu_j \geq 0)} \right] \\ &= \int_{-\infty}^{+\infty} (z^T q_i + \mu_i) \mathbb{1}_{(z^T q_i + \mu_i \geq 0)} (z^T q_j + \mu_j) \mathbb{1}_{(z^T q_j + \mu_j \geq 0)} \phi_{0,\Lambda_r}(z) dz \end{aligned} \tag{18}$$

Let's note

$$Z_{ij}^+ = \left\{ z \in \mathbb{R}^r : z^T q_i + \mu_i \geq 0,\ z^T q_j + \mu_j \geq 0 \right\}$$

$$\begin{aligned} \mathbb{E}[Y_i Y_j] &= \int_{Z_{ij}^+} z^T q_i z^T q_j \phi_{0,\Lambda_r}(z) dz \\ &+ \mu_i \int_{Z_{ij}^+} z^T q_j \phi_{0,\Lambda_r}(z) dz \\ &+ \mu_j \int_{Z_{ij}^+} z^T q_i \phi_{0,\Lambda_r}(z) dz \\ &+ \mu_i \mu_j \int_{Z_{ij}^+} \phi_{0,\Lambda_r}(z) dz \end{aligned} \tag{19}$$

$$\mathbb{E}[Y_i Y_j] = (*) + \mu_i (**)_j + \mu_j (**)_i + \mu_i \mu_j (***) \tag{20}$$

3.2.1. Special case where i=j: (Variance estimation)

$$\mathbb{E}[Y_i^2] = \int_{Z_i^+} (z^T q_i)^2 \phi_{0,\Lambda_r}(z) dz + 2\mu_i \int_{Z_i^+} z^T q_i \phi_{0,\Lambda_r}(z) dz + \mu_i^2 P(Z \in Z_i^+) \tag{21}$$

with

$$\int_{Z_i^+} (z^T q_i)^2 \phi_{0,\Lambda_r}(z) dz = c_i^2 (2\pi)^{-\frac{1}{2}} \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r^2 \exp\left(-\frac{1}{2} b_r^2 \right) db_r \tag{22}$$

Then by integration by parts we have:

$$\boxed{\mathbb{E}[Y_i^2] = (\mu_i^2 + c_i^2) \Phi_{0,1}\left(\frac{\mu_i}{c_i} \right) + \mu_i c_i \phi_{0,1}\left(\frac{\mu_i}{c_i} \right)} \tag{23}$$

3.2.2. General case where $i \neq$ j: (Covariance estimation)

We need to perform an integration over a quarter space of $\mathbb{R}^r$. By noting as previously $a = \sqrt{\Lambda_r^{-1}} z$, we have

$$Z_{ij}^+ = \left\{ a \in \mathbb{R}^r : a^T \sqrt{\Lambda_r} q_i + \mu_i \geq 0,\ a^T \sqrt{\Lambda_r} q_j + \mu_j \geq 0 \right\}$$

It is useful to note that $Z_{ij}^+ = Z_i^+ \cap Z_j^+$

Once again, by rotation we can find a unitary matrix B such that:

$$\begin{cases} B\sqrt{\Lambda_r} q_i = c_i e_r \\ B\sqrt{\Lambda_r} q_j = \alpha e_{r-1} + \beta e_r \end{cases} \tag{24}$$

From now on, we suppose that $\alpha \neq 0$. Otherwise, if $\alpha = 0$, then $\beta = c_j$ and

$$\tilde{Z}_{ij}^+ = \left\{ b \in \mathbb{R}^r : b_r \geq m = \max\left(-\frac{\mu_i}{c_i}, -\frac{\mu_j}{c_j} \right) \right\},$$

which corresponds to the previous case of integration over an half space of R^r:

$$\boxed{\mathbb{E}[Y_i Y_j] = (\mu_i \mu_j + c_i c_j)\Phi_{0,1}(-m) + (\mu_i c_j + \mu_j c_i + m c_i c_j)\phi_{0,1}(m)} \tag{25}$$

We can find the matrix B by choosing r-2 orthonormal vectors in $Z_i^+ \cap Z_j^+$, one vector in Z_i^+ and one orthogonal vector to complete the base (by Gram Schmidt for example).

We know that unitary matrix does preserve the L_2 norm and thus:

$$\begin{cases} \alpha^2 + \beta^2 = c_j^2 \\ \left\| B(\sqrt{\Lambda_r} q_i - \sqrt{\Lambda_r} q_j) \right\| = \left\| \sqrt{\Lambda_r} q_i - \sqrt{\Lambda_r} q_j \right\| \end{cases}$$

We obtain:

$$\begin{cases} \beta = \frac{c_i^2 + c_j^2 - c_{i-j}^2}{2c_i} \\ \alpha = \pm\sqrt{c_j^2 - \beta^2} \end{cases}$$

where

$$\begin{cases} c_i = \left\| \sqrt{\Lambda_r} q_i \right\| \\ c_j = \left\| \sqrt{\Lambda_r} q_j \right\| \\ c_{i-j} = \left\| \sqrt{\Lambda_r} q_i - \sqrt{\Lambda_r} q_j \right\| \end{cases}$$

WLOG, we can choose $\alpha = \sqrt{c_j^2 - \beta^2}$. We have then:

$$\begin{aligned} \tilde{Z}_{ij}^+ &= \left\{ b \in \mathbb{R}^r : b_r \geq -\frac{\mu_i}{ci}\ ,\ \alpha b_{r-1} + \beta b_r \geq -\mu_j \right\} \\ &= \left\{ b \in \mathbb{R}^r : b_r \geq -\frac{\mu_i}{ci}\ ,\ b_{r-1} \geq -\frac{(\mu_j + \beta b_r)}{\alpha} \right\} \end{aligned} \tag{26}$$

$$
\begin{aligned}
(***) &= \int_{Z_{ij}^{+}} \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \int_{-\frac{(\mu_j+\beta b_r)}{\alpha}}^{+\infty} \phi_{0,1}\left(b_r\right)\phi_{0,1}\left(b_{r-1}\right)db_{r-1}db_r \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}\left(b_r\right)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&= \int_{-\infty}^{+\infty} \phi_{0,1}\left(b_r\right)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r - \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}\left(b_r\right)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&= \Phi_{0,1}\left(\frac{\mu_j}{c_j}\right) - \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j},-\frac{\mu_i}{c_i}\right]^T\right)
\end{aligned}
\tag{27}
$$

The last equation is obtained by Owen (1980) (equations 10,010.1 and 10,010.8).

$$
\begin{aligned}
(**)_i &= \int_{Z_{ij}^{+}} z^T q_i \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \int_{-\frac{(\mu_j+\beta b_r)}{\alpha}}^{+\infty} c_i b_r \phi_{0,1}\left(b_r\right)\phi_{0,1}\left(b_{r-1}\right)db_{r-1}db_r \\
&= c_i \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r \phi_{0,1}\left(b_r\right)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r
\end{aligned}
\tag{28}
$$

By integration by parts:

$$
(**)_i = c_i\left[\left[-\phi_{0,1}\left(b_r\right)\Phi_{0,1}\left(\tfrac{\mu_j+\beta b_r}{\alpha}\right)\right]_{-\frac{\mu_i}{c_i}}^{+\infty} + \tfrac{\beta}{\alpha}\int_{-\frac{\mu_i}{c_i}}^{+\infty}\phi_{0,1}(b_r)\phi_{0,1}(\tfrac{\mu_j+\beta b_r}{\alpha})db_r\right]
\tag{29}
$$

with

$$
\phi_{0,1}\left(b_r\right)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) = \phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\phi_{0,1}\left(\frac{c_j}{\alpha}b_r + \frac{\beta\mu_j}{c_j\alpha})\right)
$$

and

$$
\begin{aligned}
\int_{-A}^{+\infty} \phi_{0,1}\left(ax+b\right)dx &= \frac{1}{a}\int_{-aA+b}^{\mathrm{sgn}(a)\infty} \phi_{0,1}\left(X\right)dX \\
&= \frac{\mathrm{sgn}(a)}{a}\Phi_{0,1}(\mathrm{sgn}(a)(aA-b))
\end{aligned}
\tag{30}
$$

Then

$$(**)_i = c_i \left[\phi_{0,1}\left(\frac{\mu_i}{c_i}\right) \Phi_{0,1}\left(\frac{\mu_j - \beta\frac{\mu_i}{c_i}}{\alpha}\right) + \frac{\beta}{\alpha}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\frac{c_j}{\alpha}\Phi_{0,1}\left(\left(\frac{c_j\mu_i}{c_i\alpha} - \frac{\beta\mu_j}{\alpha c_j}\right)\right)\right] \tag{31}$$

$$= c_i\phi_{0,1}\left(\frac{\mu_i}{c_i}\right) \Phi_{0,1}\left(\frac{\mu_j - \beta\frac{\mu_i}{c_i}}{\alpha}\right) + \frac{\beta c_i}{c_j}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\Phi_{0,1}\left(\left(\frac{c_j\mu_i}{c_i\alpha} - \frac{\beta\mu_j}{\alpha c_j}\right)\right) \tag{32}$$

$$\begin{aligned}
(**)_j &= \int_{Z_{ij}^+} z^T q_j \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty}\int_{-\frac{(\mu_j+\beta b_r)}{\alpha}}^{+\infty} (\alpha b_{r-1} + \beta b_r)\phi_{0,1}(b_r)\,\phi_{0,1}(b_{r-1})\,db_{r-1}db_r \\
&= \alpha\int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}(b_r)\,\phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right)db_r + \beta\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}(b_r)\,\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&= \alpha\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\frac{\mu_i}{c_i}}^{+\infty}\phi_{0,1}\left(\frac{c_j}{\alpha}\left(br + \frac{\beta\mu_j}{c_j^2}\right)\right)db_r + \frac{\beta}{c_i}(**)_i \\
&= c_j\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\Phi_{0,1}\left(\left(\frac{c_j\mu_i}{c_i\alpha} - \frac{\beta\mu_j}{\alpha c_j}\right)\right) + \beta\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi_{0,1}\left(\frac{\mu_j}{\alpha} - \frac{\mu_i\beta}{c_i\alpha}\right)
\end{aligned} \tag{33}$$

$$\begin{aligned}
(*) &= \int_{Z_{ij}^+} z^T q_i z^T q_j \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty}\int_{-\frac{(\mu_j+\beta b_r)}{\alpha}}^{+\infty} (c_i b_r)(\alpha b_{r-1} + \beta b_r)\phi_{0,1}(b_r)\,\phi_{0,1}(b_{r-1})\,db_{r-1}db_r \\
&= \beta c_i\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r^2\phi_{0,1}(b_r)\int_{-\frac{(\mu_j+\beta b_r)}{\alpha}}^{+\infty}\phi_{0,1}(b_{r-1})\,db_{r-1}db_r \\
&+ \alpha c_i\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}(b_r)\int_{-\frac{(\mu_j+\beta b_r)}{\alpha}}^{+\infty} b_{r-1}\phi_{0,1}(b_{r-1})\,db_{r-1}db_r \\
&= \beta c_i\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r^2\phi_{0,1}(b_r)\,\Phi_{0,1}\left(\frac{1}{\alpha}(\mu_j+\beta b_r)\right)db_r \\
&+ \alpha c_i\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}(b_r)\,\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r
\end{aligned} \tag{34}$$

the second term of the sum in equation (34) can be rewritten as:

$$\begin{aligned}
&\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}(b_r)\,\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&= \phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}\left(\frac{c_j}{\alpha}b_r + \frac{\beta\mu_j}{c_j\alpha}\right)db_r \\
&= \phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\left[\frac{\alpha^2}{c_i^2}\phi_{0,1}\left(\frac{\beta\mu_j}{\alpha c_j} - \frac{\mu_i c_j}{c_i\alpha}\right) - \frac{\alpha\beta\mu_j}{c_j^3}\Phi_{0,1}\left(\left(\frac{c_j\mu_i}{c_i\alpha} - \frac{\beta\mu_j}{\alpha c_j}\right)\right)\right]
\end{aligned} \tag{35}$$

and the first term as:

$$
\begin{aligned}
& \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r^2 \phi_{0,1}(b_r)\, \Phi_{0,1}\left(\frac{1}{\alpha}(\mu_j + \beta b_r)\right) db_r = \\
= & \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r \phi_{0,1}(b_r)\, b_r \Phi_{0,1}\left(\frac{1}{\alpha}(\mu_j + \beta b_r)\right) db_r \\
= & \left[-b_r \phi_{0,1}(b_r)\, \Phi_{0,1}\left(\frac{\mu_j + \beta b_r}{\alpha}\right)\right]_{-\frac{\mu_i}{c_i}}^{+\infty} \\
+ & \int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}(b_r)\, \Phi_{0,1}\left(\frac{1}{\alpha}(\mu_j + \beta b_r)\right) db_r \\
+ & \frac{\beta}{\alpha} \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r \phi_{0,1}(b_r)\, \phi_{0,1}\left(\frac{1}{\alpha}(\mu_j + \beta b_r)\right) db_r \\
= & -\frac{\mu_i}{c_i} \phi_{0,1}\left(\frac{\mu_i}{c_i}\right) \Phi\left(\frac{1}{\alpha}\left(\mu_j - \beta \frac{\mu_i}{c_i}\right)\right) + \Phi_{0,1}\left(\frac{\mu_j}{c_j}\right) - \Phi_{ij} \\
+ & \frac{\beta}{\alpha} \phi_{0,1}\left(\frac{\mu_j}{c_j}\right) \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r \phi_{0,1}\left(\frac{c_j}{\alpha}\left(b_r + \frac{\beta \mu_j}{c_j^2}\right)\right) db_r
\end{aligned} \tag{36}
$$

Recombining the two terms of equations (35)- (36) yields to:

$$
\begin{aligned}
(*) \quad = \quad & c_i \alpha \phi_{0,1}\left(\frac{\mu_j}{c_j}\right) \phi_{0,1}\left(\frac{\beta \mu_j}{\alpha c_j} - \frac{\mu_i c_j}{c_i \alpha}\right) \\
- \quad & \phi_{0,1}\left(\frac{\mu_j}{c_j}\right) \frac{c_i \beta \mu_j}{c_j} \Phi_{0,1}\left(\left(\frac{c_j \mu_i}{c_i \alpha} - \frac{\beta \mu_j}{\alpha c_j}\right)\right) \\
+ \quad & c_i \beta (***) \\
- \quad & \beta \mu_i \phi_{0,1}\left(\frac{\mu_i}{c_i}\right) \Phi_{0,1}\left(\frac{\mu_j}{\alpha} - \frac{\beta \mu_i}{\alpha c_i}\right)
\end{aligned} \tag{37}
$$

Using the following notations:

$$
\phi_j = \phi_{0,1}\left(\frac{\mu_j}{c_j}\right), \tag{38}
$$

$$
\phi_i = \phi_{0,1}\left(\frac{\mu_i}{c_i}\right), \tag{39}
$$

$$
\Phi_j = \Phi_{0,1}\left(\frac{\mu_j}{c_j}\right), \tag{40}
$$

$$
\Phi_{ij} = \Phi_{\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j}, -\frac{\mu_i}{c_i}\right]^T\right), \tag{41}
$$

$$
\Phi_{NS} = \Phi_{0,1}\left(\frac{\mu_j}{\alpha} - \frac{\beta \mu_i}{\alpha c_i}\right), \tag{42}
$$

$$
\Phi_S = \Phi_{0,1}\left(\left(\frac{c_j \mu_i}{c_i \alpha} - \frac{\beta \mu_j}{\alpha c_j}\right)\right), \tag{43}
$$

$$
\phi_{NS} = \phi_{0,1}\left(\frac{\beta \mu_j}{\alpha c_j} - \frac{\mu_i c_j}{c_i \alpha}\right), \tag{44}
$$

we obtain:

$$\begin{aligned}
\mathbb{E}[Y_iY_j] &= c_i\alpha\phi_j\phi_{NS} \\
&- \frac{c_i\beta\mu_j}{c_j}\phi_j\Phi_S \\
&+ (\mu_i\mu_j + c_i\beta)(\Phi_j - \Phi_{ij}) \\
&- \beta\mu_i\phi_i\Phi_{NS} \\
&+ \mu_i c_j\phi_j\Phi_S \\
&+ \beta\mu_i\phi_i\Phi_{NS} \\
&+ \mu_j c_i\phi_i\Phi_{NS} \\
&+ \frac{c_i\beta\mu_j}{c_j}\phi_j\Phi_S
\end{aligned} \tag{45}$$

$$\boxed{\mathbb{E}[Y_iY_j] = c_i\alpha\phi_j\phi_{NS} + (\mu_i\mu_j + c_i\beta)(\Phi_j - \Phi_{ij}) + \mu_i c_j\phi_j\Phi_S + \mu_j c_i\phi_i\Phi_{NS}} \tag{46}$$

4. Derivation for Leaky Relu

4.1. Expected value estimation

Let's note f_i such that:

$$\begin{aligned}
f_i &: \mathbb{R}^n \to \mathbb{R}_+ \\
x &\mapsto \begin{cases} x_i \text{ if } x_i \geq 0 \\ \lambda x_i \text{ otherwise, } \lambda \in [0,1] \end{cases}
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}[Y_i] &= \mathbb{E}_{z\sim p(Z)}\left[f_i(Q_r z + \mu)\right] \\
&= \int_{-\infty}^{+\infty} f_i(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz
\end{aligned} \tag{47}$$

Let's note

$$Z_i^+ = \left\{z \in \mathbb{R}^r : z^T q_i + \mu_i \geq 0\right\},$$

and:

$$Z_i^- = \left\{z \in \mathbb{R}^r : z^T q_i + \mu_i < 0\right\},$$

then

$$\begin{aligned}
\mathbb{E}[Y_i] &= \int_{Z_i^+} f_i(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz + \int_{Z_i^-} f_i(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \\
&= \int_{Z_i^+} (z^T q_i + \mu_i)\phi_{0,\Lambda_r}(z)dz + \lambda\int_{Z_i^-} (z^T q_i + \mu_i)\phi_{0,\Lambda_r}(z)dz
\end{aligned} \tag{48}$$

Yet, from the equation (17), we have:

$$\int_{Z_i^+} (z^T q_i + \mu_i)\phi_{0,\Lambda_r}(z)dz = c_i\phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \mu_i\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)$$

and:

$$\begin{aligned}
\int_{Z_i^-} (z^T q_i + \mu_i)\phi_{0,\Lambda_r}(z)dz &= \int_{Z_i^-} (z^T q_i)\phi_{0,\Lambda_r}(z)dz + \mu_i \int_{Z_i^-} \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} c_i b_r \phi_{0,1}(b_r)db_r + \mu_i \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)db_r \\
&= c_i\left[\left[-\phi_{0,1}(x)\right]_{-\infty}^{-\frac{\mu_i}{c_i}}\right] + \mu_i\Phi_{0,1}\left(-\frac{\mu_i}{c_i}\right) \\
&= -c_i\phi_{0,1}(\frac{\mu_i}{c_i}) + \mu_i\left(1 - \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\right) \\
&= -\left(c_i\phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \mu_i\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\right) + \mu_i
\end{aligned} \tag{49}$$

Thus, we obtain:

$$\boxed{\mathbb{E}[Y_i] = (1-\lambda)c_i\phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + (1-\lambda)\mu_i\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \lambda\mu_i} \tag{50}$$

4.2. Variance value estimation

$$\begin{aligned}
\mathbb{E}[Y_i^2] &= \mathbb{E}_{z\sim p(Z)}\left[f_i(Q_r z + \mu)^2\right] \\
&= \int_{-\infty}^{+\infty} f_i(Q_r z + \mu)^2 \phi_{0,\Lambda_r}(z)dz \\
&= \int_{Z_i^+} (z^T q_i + \mu_i)^2 \phi_{0,\Lambda_r}(z)dz + \lambda^2 \int_{Z_i^-} (z^T q_i + \mu_i)^2 \phi_{0,\Lambda_r}(z)dz
\end{aligned}$$

Yet, by the equation (23):

$$\int_{Z_i^+} (z^T q_i + \mu_i)^2 \phi_{0,\Lambda_r}(z)dz = (\mu_i^2 + c_i^2)\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \mu_i c_i \phi_{0,1}\left(\frac{\mu_i}{c_i}\right)$$

Now:

$$\begin{aligned}
\int_{Z_i^-} (z^T q_i + \mu_i)^2 \phi_{0,\Lambda_r}(z)dz = \int_{Z_i^-} (z^T q_i)^2 \phi_{0,\Lambda_r}(z)dz &+ 2\mu_i \int_{Z_i^-} z^T q_i \phi_{0,\Lambda_r}(z)dz \\
&+ \mu_i^2 \int_{Z_i^-} \phi_{0,\Lambda_r}(z)dz
\end{aligned}$$

with:

$$\int_{Z_i^-} \phi_{0,\Lambda_r}(z)dz = 1 - \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) \tag{51}$$

$$\int_{Z_i^-} z^T q_i \phi_{0,\Lambda_r}(z)dz = -c_i\phi_{0,1}\left(\frac{\mu_i}{c_i}\right) \tag{52}$$

$$\begin{aligned}\int_{Z_i^-} (z^T q_i)^2 + &= c_i^2 \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r^2 \phi_{0,1}(b_r)db_r \\ &= c_i^2 \left([-x\phi_{0,1}(x)]_{-\infty}^{-\frac{\mu_i}{c_i}} + \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)db_r \right) \\ &= \mu_i c_i \phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + c_i^2\left(1 - \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\right)\end{aligned} \tag{53}$$

Then:

$$\int_{Z_i^-} (z^T q_i + \mu_i)^2 \phi_{0,\Lambda_r}(z)dz = -\left((\mu_i^2 + c_i^2)\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \mu_i c_i \phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\right) + c_i^2 + \mu_i^2 \tag{54}$$

We finally obtain:

$$\boxed{\mathbb{E}[Y_i^2] = (1-\lambda^2)(\mu_i^2 + c_i^2)\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + (1-\lambda^2)\mu_i c_i \phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \lambda^2(c_i^2 + \mu_i^2)} \tag{55}$$

4.3. Covariance value estimation

Let's note f_{ij} such that:

$$\begin{aligned} f_{ij} \colon \mathbb{R}^n &\to \mathbb{R}_+ \\ x &\mapsto f_i(x) f_j(x) \end{aligned}$$

with f_i defined as in (47).

$$\begin{aligned}\mathbb{E}[Y_i Y_j] &= \mathbb{E}_{z\sim p(Z)}\left[f_{ij}(Q_r z + \mu)\right] \\ &= \mathbb{E}_{z\sim p(Z)}\left[f_i(Q_r z + \mu) f_j(Q_r z + \mu)\right]\end{aligned} \tag{56}$$

By noting:

$$\begin{aligned} Z_{ij}^+ &= \left\{z \in \mathbb{R}^r : z^T q_i + \mu_i \geq 0,\ z^T q_j + \mu_j \geq 0\right\} \\ Z_{ij}^- &= \left\{z \in \mathbb{R}^r : z^T q_i + \mu_i < 0,\ z^T q_j + \mu_j < 0\right\} \\ Z_{ij}^{-+} &= \left\{z \in \mathbb{R}^r : z^T q_i + \mu_i \geq 0,\ z^T q_j + \mu_j < 0\right\} \end{aligned}$$

$$Z_{ij}^{+-} = \left\{ z \in \mathbb{R}^r : z^T q_i + \mu_i < 0,\ z^T q_j + \mu_j \geq 0 \right\}$$

We have:

$$\begin{aligned}
\mathbb{E}[Y_i Y_j] &= \int_{z_{ij}^{+}} f_{ij}(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \\
&+ \int_{z_{ij}^{-}} f_{ij}(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \\
&+ \int_{z_{ij}^{+-}} f_{ij}(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \\
&+ \int_{z_{ij}^{-+}} f_{ij}(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \\
&= (+) + (-) + (+-) + (-+)
\end{aligned} \tag{57}$$

4.3.1. DERIVATION OVER Z_{ij}^{-}:

$$\begin{aligned}
(-) &= \int_{z_{ij}^{-}} f_{ij}(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \\
&= \lambda^2 \int_{z_{ij}^{-}} (z^T q_i + \mu_i)(z^T q_j + \mu_j)\phi_{0,\Lambda_r}(z)dz \\
&= \lambda^2 [\int_{z_{ij}^{-}} z^T q_i z^T q_j \phi_{0,\Lambda_r}(z)dz + \mu_j \int_{z_{ij}^{-}} z^T q_i \phi_{0,\Lambda_r}(z)dz + \mu_i \int_{z_{ij}^{-}} z^T q_j \phi_{0,\Lambda_r}(z)dz \\
&+ \mu_i \mu_j \int_{z_{ij}^{-}} \phi_{0,\Lambda_r}(z)dz] \\
&= \lambda^2 [(-*) + \mu_j(-**)_i + \mu_i(-**)_j + \mu_i \mu_j(-***)]
\end{aligned} \tag{58}$$

With:

$$\begin{aligned}
(-***) &= \int_{z_{ij}^{-}} \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\infty}^{-\frac{\mu_j + \beta b_r}{\alpha}} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1})db_r db_{r-1} \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)\Phi_{0,1}(-\frac{\mu_j + \beta b_r}{\alpha})db_r \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{01}(b_r)db_r - \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)\Phi_{0,1}(\frac{\mu_j + \beta b_r}{\alpha})db_r \\
&= \Phi_{0,1}\left(-\frac{\mu_i}{c_i}\right) - \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j}, -\frac{\mu_i}{c_i}\right]^T\right) \\
&= 1 - \Phi_i - \Phi_{ij}
\end{aligned} \tag{59}$$

and:

$$
\begin{aligned}
(-**)_i &= \int_{z_{ij}^-} z^T q_i \phi_{0,\Lambda_r}(z) dz \\
&= c_i \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} b_r \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= c_i \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= c_i [\left[-\phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right)\right]_{-\infty}^{-\frac{\mu_i}{c_i}} \\
&- \frac{\beta}{\alpha}\int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r] \\
&= c_i [-\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi_{0,1}\left(-\frac{\mu_j}{\alpha}+\frac{\beta\mu_i}{\alpha c_i}\right) \\
&- \frac{\beta}{\alpha}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r] \\
&= -c_i\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi_{0,1}\left(-\frac{\mu_j}{\alpha}+\frac{\beta\mu_i}{\alpha c_i}\right) - \frac{c_i\beta}{c_j}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\Phi_{0,1}\left(\frac{\beta\mu_j}{\alpha c_j}-\frac{c_j\mu_i}{\alpha c_i}\right) \\
&= -c_i\phi_i(1-\Phi_{NS}) - \frac{c_i\beta}{c_j}\phi_j(1-\Phi_S) \qquad (60)
\end{aligned}
$$

and:

$$
\begin{aligned}
(-**)_j &= \int_{z_{ij}^-} z^T q_j \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} (\alpha b_{r-1}+\beta b_r)\phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= \beta \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&- \alpha \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= \beta \left[-\phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right)\right]_{-\infty}^{-\frac{\mu_i}{c_i}} \\
&- (\frac{\beta^2}{\alpha}+\alpha)\int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= -\beta\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi_{0,1}\left(-\frac{\mu_j}{\alpha}+\frac{\beta\mu_i}{\alpha c_i}\right) - \frac{c_j^2}{\alpha}\int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= -\beta\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi_{0,1}\left(\frac{\beta\mu_i}{\alpha c_i}-\frac{\mu_j}{\alpha}\right) - c_j\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\Phi_{0,1}\left(\frac{\beta\mu_j}{\alpha c_j}-\frac{c_j\mu_i}{\alpha c_i}\right) \\
&= -\beta\phi_i(1-\Phi_{NS}) - c_j\phi_j(1-\Phi_S) \qquad (61)
\end{aligned}
$$

and:

$$
\begin{aligned}
(-*) &= \int_{z_{ij}^-} z^T q_i z^T q_j \phi_{0,\Lambda_r}(z) dz \\
&= c_i \beta \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} b_r^2 \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&+ \alpha c_i \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} b_{r-1} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= c_i \beta \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r^2 \phi_{0,1}(b_r) \Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&- \alpha c_i \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_{r-1} \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r
\end{aligned} \tag{62}
$$

With:

$$
\begin{aligned}
&\int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r = \phi_{0,1}\left(\frac{\mu_j}{c_j}\right) \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r \phi_{0,1}\left(\frac{c_j}{\alpha} b_r + \frac{\beta \mu_j}{\alpha c_j}\right) db_r \\
= \;& \phi_{0,1}\left(\frac{\mu_j}{c_j}\right) \left[\frac{\alpha^2}{c_j^2} \int_{-\infty}^{-\frac{c_j\mu_i}{\alpha c_i}+\frac{\beta\mu_j}{\alpha c_j}} x\phi_{0,1}(x)dx - \frac{\alpha\beta\mu_j}{c_j^3} \int_{-\infty}^{-\frac{c_j\mu_i}{\alpha c_i}+\frac{\beta\mu_j}{\alpha c_j}} \phi_{0,1}(x)dx \right] \\
= \;& -\frac{\alpha^2}{c_j^2} \phi_{0,1}\left(\frac{\mu_j}{c_j}\right) \phi_{0,1}\left(\frac{\beta\mu_j}{\alpha c_j} - \frac{c_j\mu_i}{\alpha c_i}\right) - \frac{\alpha\beta\mu_j}{c_j^3} \phi_{0,1}\left(\frac{\mu_j}{c_j}\right) \Phi_{0,1}\left(\frac{\beta\mu_j}{\alpha c_j} - \frac{c_j\mu_i}{\alpha c_i}\right) \\
= \;& -\frac{\alpha^2}{c_j^2} \phi_j \phi_{NS} - \frac{\alpha\beta\mu_j}{c_j^3} \phi_j (1 - \Phi_S)
\end{aligned} \tag{63}
$$

and:

$$
\begin{aligned}
& \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r^2 \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r = \left[-b_r\phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right)\right]_{-\infty}^{-\frac{\mu_i}{c_i}} \\
+ & \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r - \frac{\beta}{\alpha}\int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r\phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
= & \frac{\mu_i}{c_i}\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi_{0,1}\left(-\frac{\mu_j}{\alpha}+\frac{\beta\mu_i}{\alpha c_i}\right) + (-**) + \frac{\beta\alpha}{c_j^2}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\phi_{0,1}\left(-\frac{c_j\mu_i}{\alpha c_i}+\frac{\beta\mu_j}{\alpha c_j}\right) \\
+ & \frac{\beta^2\mu_j}{c_j^3}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\Phi_{0,1}\left(-\left(\frac{c_j\mu_i}{\alpha c_i}-\frac{\beta\mu_j}{\alpha c_j}\right)\right) \\
= & \frac{\mu_i}{c_i}\phi_i(1-\Phi_{NS}) + (-**) + \frac{\beta\alpha}{c_j^2}\phi_j\phi_{NS} + \frac{\beta^2\mu_j}{c_j^3}\phi_j(1-\Phi_S)
\end{aligned} \tag{64}
$$

Finally:

$$
(-*) = \beta\mu_i\phi_i(1-\Phi_{NS}) + \beta c_i(-**) + \alpha c_i\phi_j\phi_{NS} + \frac{\beta\mu_j c_i}{c_j}\phi_j(1-\Phi_S) \tag{65}
$$

Then:

$$
\begin{aligned}
(-) = & \lambda^2[\beta\mu_i\phi_i(1-\Phi_{NS}) + (\beta c_i+\mu_i\mu_j)(1-\Phi_i-\Phi_{ij}) + \alpha c_i\phi_j\phi_{NS} \\
+ & \frac{\beta\mu_j c_i}{c_j}\phi_j(1-\Phi_S) - \mu_i\beta\phi_i(1-\Phi_{NS}) - \mu_i c_j\phi_j(1-\Phi_S) - \mu_j c_i\phi_i(1-\Phi_{NS}) \\
- & \frac{c_i\beta\mu_j}{c_j}\phi_j(1-\Phi_S)] \\
= & \lambda^2[(\beta c_i+\mu_i\mu_j)(1-\Phi_i-\Phi_{ij}) + \alpha c_i\phi_j\phi_{NS} - \mu_j c_i\phi_i(1-\Phi_{NS}) \\
- & \mu_i c_j\phi_j(1-\Phi_S)]
\end{aligned} \tag{66}
$$

4.3.2. DERIVATION OVER Z_{ij}^{+-}:

$$
\begin{aligned}
(+-) = & \int_{z_{ij}^{+-}} f_{ij}(Q_r z+\mu)\phi_{0,\Lambda_r}(z)dz \\
= & \lambda\int_{z_{ij}^{+-}} (z^T q_i+\mu_i)(z^T q_j+\mu_j)\phi_{0,\Lambda_r}(z)dz \\
= & \lambda[\int_{z_{ij}^{+-}} z^T q_i z^T q_j\phi_{0,\Lambda_r}(z)dz + \mu_j\int_{z_{ij}^{+-}} z^T q_i\phi_{0,\Lambda_r}(z)dz \\
+ & \mu_i\int_{z_{ij}^{+-}} z^T q_j\phi_{0,\Lambda_r}(z)dz + \mu_i\mu_j\int_{z_{ij}^{+-}} \phi_{0,\Lambda_r}(z)dz] \\
= & \lambda[(+-*) + \mu_j(+-**)_i + \mu_i(+-**)_j + \mu_i\mu_j(+-***)]
\end{aligned} \tag{67}
$$

With:

$$
\begin{aligned}
(+-*) &= \int_{z_{ij}^{+-}} z^T q_i z^T q_j \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} (c_i b_r)(\alpha b_{r-1} + \beta b_r)\phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= \beta c_i \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r^2 \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&- \alpha c_i \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r
\end{aligned} \tag{68}
$$

with:

$$
\begin{aligned}
\int_{-\frac{\mu_i}{c_i}}^{+\infty} & b_r^2 \quad \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r = \left[-b_r\phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right)\right]_{-\frac{\mu_i}{c_i}}^{+\infty} \\
&+ \int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r - \frac{\beta}{\alpha}\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= -\frac{\mu_i}{c_i}\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi_{0,1}\left(-\frac{\mu_j}{\alpha}+\frac{\beta\mu_i}{\alpha c_i}\right) + (+-***) \\
&- \frac{\beta}{\alpha}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}\left(\frac{c_j}{\alpha}b_r + \frac{\beta\mu_j}{\alpha c_j}\right) db_r \\
&= -\frac{\mu_i}{c_i}\phi_i(1-\Phi_{NS}) + (+-***) - \frac{\beta\alpha}{c_j^2}\phi_j\phi_{NS} + \frac{\beta^2\mu_j}{c_j^3}\phi_j\Phi_S
\end{aligned} \tag{69}
$$

and:

$$
\begin{aligned}
\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r &= \phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}\left(\frac{c_j}{\alpha}b_r + \frac{\beta\mu_j}{\alpha c_j}\right) db_r \\
&= \frac{\alpha^2}{c_j^2}\phi_j\phi_{NS} - \frac{\alpha\beta\mu_j}{c_j^3}\phi_j\Phi_S
\end{aligned} \tag{70}
$$

$$
\tag{71}
$$

Then:

$$
\begin{aligned}
(+-*) &= -\beta\mu_i\phi_i(1-\Phi_{NS}) + \beta c_i(+-***) - \frac{\beta^2\alpha c_i}{c_j^2}\phi_j\phi_{NS} \\
&+ \frac{\beta^3 c_i\mu_j}{c_j^3}\phi_j\Phi_S - \frac{\alpha^3 c_i}{c_j^2}\phi_j\phi_{NS} + \frac{\beta\alpha^2 c_i\mu_j}{c_j^3}\phi_j\Phi_S \\
&= -\beta\mu_i\phi_i(1-\Phi_{NS}) + \beta c_i(+-***) - \alpha c_i\phi_j\phi_{NS} + \frac{\beta c_i\mu_j}{c_j}\phi_j\Phi_S
\end{aligned} \tag{72}
$$

$$
\tag{73}
$$

and:

$$
\begin{aligned}
(+-**)_i &= \int_{z_{ij}^{+-}} z^T q_i \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} c_i b_r \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} c_i b_r \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= c_i \left[\left[-\phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right)\right]_{-\frac{\mu_i}{c_i}}^{+\infty}\right] - \frac{\beta}{\alpha}\int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= c_i \phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi\left(-\frac{\mu_j}{\alpha}+\frac{\beta\mu_j}{\alpha c_i}\right) - \frac{\beta c_i}{c_j}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\Phi_{0,1}\left(\left(\frac{c_j\mu_i}{c_i\alpha}-\frac{\beta\mu_j}{\alpha c_j}\right)\right) \\
&= c_i\phi_i(1-\Phi_{NS}) - \frac{\beta c_i}{c_j}\phi_j\Phi_S
\end{aligned} \tag{74}
$$

and:

$$
\begin{aligned}
(+-**)_j &= \int_{z_{ij}^{+-}} z^T q_j \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} (\alpha b_{r-1}+\beta b_r)\phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= -\alpha \int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}(b_r)\phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&+ \beta \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r\phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= -\alpha\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}\left(\frac{c_j}{\alpha}b_r+\frac{\beta\mu_j}{\alpha c_j}\right) db_r \\
&+ \beta\left[\left[-\phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right)\right]_{-\frac{\mu_i}{c_i}}^{+\infty}\right. \\
&- \left.\frac{\beta}{\alpha}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}\left(\frac{c_j}{\alpha}b_r+\frac{\beta\mu_j}{\alpha c_j}\right) db_r\right] \\
&= \beta\phi_i(1-\Phi_{NS}) - c_j\phi_j\Phi_S
\end{aligned} \tag{75}
$$

and:

$$
\begin{aligned}
(+-***) &= \int_{z_{ij}^{+-}} \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1})db_r db_{r-1} \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}(b_r)db_r - \int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&= \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) - \int_{-\infty}^{+\infty} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&+ \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&= \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) - \Phi_{0,1}\left(\frac{\mu_j}{c_j}\right) + \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j}, -\frac{\mu_i}{c_i}\right]^T\right) \\
&= \Phi_i - \Phi_j + \Phi_{ij}
\end{aligned} \tag{76}
$$

Then:

$$
\begin{aligned}
(+-) &= \lambda[(+-*) + \mu_j(+-**)_i + \mu_i(+-**)_j + \mu_i\mu_j(+-***)] \\
&= \lambda[-\beta\mu_i\phi_i(1-\Phi_{NS}) + \beta c_i(\Phi_i - \Phi_j + \Phi_{ij}) - \alpha c_i\phi_j\phi_{NS} + \frac{\beta c_i\mu_j}{c_j}\phi_j\Phi_S \\
&+ c_i\mu_j\phi_i(1-\Phi_{NS}) - \frac{\beta c_i\mu_j}{c_j}\phi_j\Phi_S \\
&+ \beta\mu_i\phi_i(1-\Phi_{NS}) - c_j\mu_i\phi_j\Phi_S \\
&+ \mu_i\mu_j(\Phi_i - \Phi_j + \Phi_{ij})] \\
&= \lambda[(\beta c_i + \mu_i\mu_j)(\Phi_i - \Phi_j + \Phi_{ij}) - \alpha c_i\phi_j\phi_{NS} - c_j\mu_i\phi_j\Phi_S \\
&+ c_i\mu_j\phi_i(1-\Phi_{NS})]
\end{aligned} \tag{77}
$$

4.3.3. DERIVATION OVER Z_{ij}^{-+}:

$$
\begin{aligned}
(-+) &= \int_{z_{ij}^{-+}} f_{ij}(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \\
&= \lambda \int_{z_{ij}^{-+}} (z^T q_i + \mu_i)(z^T q_j + \mu_j)\phi_{0,\Lambda_r}(z)dz \\
&= \lambda[\int_{z_{ij}^{-+}} z^T q_i z^T q_j \phi_{0,\Lambda_r}(z)dz + \mu_j \int_{z_{ij}^{-+}} z^T q_i \phi_{0,\Lambda_r}(z)dz \\
&+ \mu_i \int_{z_{ij}^{-+}} z^T q_j \phi_{0,\Lambda_r}(z)dz + \mu_i\mu_j \int_{z_{ij}^{-+}} \phi_{0,\Lambda_r}(z)dz] \\
&= \lambda[(-+*) + \mu_j(-+**)_i + \mu_i(-+**)_j + \mu_i\mu_j(-+***)]
\end{aligned} \tag{78}
$$

With:

$$
\begin{aligned}
(-+*) &= \int_{z_{ij}^{-+}} z^T q_i z^T q_j \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\frac{\mu_j+\beta b_r}{\alpha}}^{+\infty} (c_i b_r)(\alpha b_{r-1} + \beta b_r)\phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= \beta c_i \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r^2 \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&+ \alpha c_i \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r
\end{aligned} \tag{79}
$$

with:

$$
\begin{aligned}
\int_{-\infty}^{-\frac{\mu_i}{c_i}} & b_r^2 \quad \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r = \left[-b_r\phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)\right]_{-\infty}^{-\frac{\mu_i}{c_i}} \\
&+ \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r + \frac{\beta}{\alpha}\int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r\phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= \frac{\mu_i}{c_i}\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi_{0,1}\left(\frac{\mu_j}{\alpha} - \frac{\beta\mu_i}{\alpha c_i}\right) + (-+***) \\
&+ \frac{\beta}{\alpha}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r\phi_{0,1}\left(\frac{c_j}{\alpha}b_r + \frac{\beta\mu_j}{\alpha c_j}\right) db_r \\
&= \frac{\mu_i}{c_i}\phi_i\Phi_{NS} + (-+***) - \frac{\beta\alpha}{c_j^2}\phi_j\phi_{NS} - \frac{\beta^2\mu_j}{c_j^3}\phi_j(1-\Phi_S)
\end{aligned} \tag{80}
$$

and:

$$
\begin{aligned}
\int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r\phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r &= \phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r\phi_{0,1}\left(\frac{c_j}{\alpha}b_r + \frac{\beta\mu_j}{\alpha c_j}\right) db_r \\
&= -\frac{\alpha^2}{c_j^2}\phi_j\phi_{NS} - \frac{\alpha\beta\mu_j}{c_j^3}\phi_j(1-\Phi_S)
\end{aligned} \tag{81}
$$

Then:

$$
\begin{aligned}
(-+*) &= \beta\mu_i\phi_i\Phi_{NS} + \beta c_i(-+***) - \frac{\beta^2\alpha c_i}{c_j^2}\phi_j\phi_{NS} \\
&- \frac{\beta^3 c_i\mu_j}{c_j^3}\phi_j(1-\Phi_S) - \frac{\alpha^3 c_i}{c_j^2}\phi_j\phi_{NS} - \frac{\beta\alpha^2 c_i\mu_j}{c_j^3}\phi_j(1-\Phi_S) \\
&= \beta\mu_i\phi_i\Phi_{NS} + \beta c_i(-+***) - \alpha c_i\phi_j\phi_{NS} - \frac{\beta c_i\mu_j}{c_j}\phi_j(1-\Phi_S)
\end{aligned} \tag{82}
$$

and:

$$
\begin{aligned}
(-+**)_i &= \int_{z_{ij}^{-+}} z^T q_i \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\frac{\mu_j+\beta b_r}{\alpha}}^{+\infty} c_i b_r \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} c_i b_r \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= c_i \left[-\phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)\right]_{+\infty}^{-\frac{\mu_i}{c_i}} \\
&+ c_i \frac{\beta}{\alpha} \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= -c_i \phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\Phi\left(\frac{\mu_j}{\alpha} - \frac{\beta\mu_j}{\alpha c_i}\right) + \frac{\beta c_i}{c_j}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\Phi_{0,1}\left(\frac{\beta\mu_j}{\alpha c_j} - \frac{c_j\mu_i}{c_i\alpha}\right) \\
&= -c_i\phi_i\Phi_{NS} + \frac{\beta c_i}{c_j}\phi_j(1-\Phi_S)
\end{aligned} \tag{83}
$$

and:

$$
\begin{aligned}
(-+**)_j &= \int_{z_{ij}^{-+}} z^T q_j \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\frac{\mu_j+\beta b_r}{\alpha}}^{+\infty} (\alpha b_{r-1} + \beta b_r)\phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= \alpha \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&+ \beta \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r\phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right) db_r \\
&= \alpha\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}\left(\frac{c_j}{\alpha}b_r + \frac{\beta\mu_j}{\alpha c_j}\right) db_r \\
&+ \beta\Big[\left[-\phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)\right]_{-\infty}^{-\frac{\mu_i}{c_i}} \\
&+ \frac{\beta}{\alpha}\phi_{0,1}\left(\frac{\mu_j}{c_j}\right)\int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}\left(\frac{c_j}{\alpha}b_r + \frac{\beta\mu_j}{\alpha c_j}\right) db_r\Big] \\
&= -\beta\phi_i\Phi_{NS} + c_j\phi_j(1-\Phi_S)
\end{aligned} \tag{84}
$$

and:

$$
\begin{aligned}
(-+***) &= \int_{z_{ij}^{-+}} \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\frac{\mu_j+\beta b_r}{\alpha}}^{+\infty} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1})db_r db_{r-1} \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&= \Phi_{ij}
\end{aligned} \tag{85}
$$

Then:

$$
\begin{aligned}
(-+) &= \lambda[(-+*) + \mu_j(-+**)_i + \mu_i(-+**)_j + \mu_i\mu_j(-+***)] \\
&= \lambda[\beta\mu_i\phi_i\Phi_{NS} + \beta c_i\Phi_{ij} - \alpha c_i\phi_j\phi_{NS} - \frac{\beta c_i\mu_j}{c_j}\phi_j(1-\Phi_S) \\
&- c_i\mu_j\phi_i\Phi_{NS} + \frac{\beta c_i\mu_j}{c_j}\phi_j(1-\Phi_S) \\
&- \beta\mu_i\phi_i\Phi_{NS} + c_j\mu_i\phi_j(1-\Phi_S) \\
&+ \mu_i\mu_j\Phi_{ij}] \\
&= \lambda[(\beta c_i + \mu_i\mu_j)\Phi_{ij} - \alpha c_i\phi_j\phi_{NS} + c_j\mu_i\phi_j(1-\Phi_S) - c_i\mu_j\phi_i\Phi_{NS}]
\end{aligned} \tag{86}
$$

4.3.4. Final derivation:

Finally, by the equation (46), we have:

$$
(+) = \mathbb{E}[Y_iY_j] = c_i\alpha\phi_j\phi_{NS} + (\mu_i\mu_j + c_i\beta)(\Phi_j - \Phi_{ij}) + \mu_i c_j\phi_j\Phi_S + \mu_j c_i\phi_i\Phi_{NS}
$$

Then:

$$
\begin{aligned}
\mathbb{E}[Y_iY_j] &= (+) + (-) + (+-) + (-+) \\
&= (\beta c_i + \mu_i\mu_j)(\Phi_j - \Phi_{ij}) + c_i\alpha\phi_j\phi_{NS} + \mu_i c_j\phi_j\Phi_S + \mu_j c_i\phi_i\Phi_{NS} \\
&+ \lambda^2(\beta c_i + \mu_i\mu_j)(1 - \Phi_i - \Phi_{ij}) + \lambda^2\alpha c_i\phi_j\phi_{NS} - \lambda^2\mu_j c_i\phi_i(1-\Phi_{NS}) \\
&- \lambda^2\mu_i c_j\phi_j(1-\Phi_S) \\
&+ \lambda(\beta c_i + \mu_i\mu_j)(\Phi_i - \Phi_j + \Phi_{ij}) - \lambda\alpha c_i\phi_j\phi_{NS} - \lambda c_j\mu_i\phi_j\Phi_S + \lambda c_i\mu_j\phi_i(1-\Phi_{NS}) \\
&+ \lambda(\beta c_i + \mu_i\mu_j)\Phi_{ij} - \lambda\alpha c_i\phi_j\phi_{NS} + \lambda c_j\mu_i\phi_j(1-\Phi_S) - \lambda c_i\mu_j\phi_i\Phi_{NS}
\end{aligned} \tag{87}
$$

Finally, we obtain:

$$
\begin{aligned}
\mathbb{E}[Y_iY_j] &= (1-\lambda)^2 c_i\alpha\phi_j\phi_{NS} \\
&+ (\mu_i\mu_j + c_i\beta)\left((1-\lambda)\Phi_j - (1-\lambda)^2\Phi_{ij} + \lambda(1-\lambda)\Phi_i + \lambda^2\right) \\
&+ (1-\lambda)^2\mu_i c_j\phi_j\Phi_S \\
&+ (1-\lambda)^2\mu_j c_i\phi_i\Phi_{NS} \\
&+ \lambda(1-\lambda)(\mu_j c_i\phi_i + \mu_i c_j\phi_j)
\end{aligned} \tag{88}
$$

5. Derivation for Elu

5.1. Expected value estimation

Let's note f_i such that:

$$\begin{aligned} f_i \colon \mathbb{R}^n &\to \mathbb{R}_+ \\ x &\mapsto \begin{cases} x_i \text{ if } x_i \geq 0 \\ \lambda\left(e^{x_i} - 1\right) \text{ otherwise, } \lambda \in [0,1] \end{cases} \end{aligned}$$

$$\begin{aligned} \mathbb{E}[Y_i] &= \mathbb{E}_{z\sim p(Z)}\left[f_i(Q_r z + \mu)\right] \\ &= \int_{-\infty}^{+\infty} f_i(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \end{aligned} \tag{89}$$

Let's note

$$Z_i^+ = \left\{z \in \mathbb{R}^r : z^T q_i + \mu_i \geq 0\right\},$$

and:

$$Z_i^- = \left\{z \in \mathbb{R}^r : z^T q_i + \mu_i < 0\right\},$$

then

$$\begin{aligned} \mathbb{E}[Y_i] &= \int_{Z_i^+} f_i(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz + \int_{Z_i^-} f_i(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \\ &= \int_{Z_i^+} (z^T q_i + \mu_i)\phi_{0,\Lambda_r}(z)dz + \lambda \int_{Z_i^-} (e^{z^T q_i + \mu_i} - 1)\phi_{0,\Lambda_r}(z)dz \end{aligned} \tag{90}$$

Yet, from the equation (17), we have:

$$\int_{Z_i^+} (z^T q_i + \mu_i)\phi_{0,\Lambda_r}(z)dz = c_i \phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \mu_i \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)$$

and:

$$\begin{aligned} \int_{Z_i^-} (e^{z^T q_i + \mu_i} - 1)\phi_{0,\Lambda_r}(z)dz &= e^{\mu_i} \int_{Z_i^-} e^{z^T q_i}\phi_{0,\Lambda_r}(z)dz - \int_{Z_i^-} \phi_{0,\Lambda_r}(z)dz \\ &= e^{\mu_i} \int_{-\infty}^{-\frac{\mu_i}{c_i}} e^{c_i b_r}\phi_{0,1}(b_r)db_r - \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r)db_r \\ &= e^{\mu_i + \frac{c_i^2}{2}} \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r - c_i)db_r - (1 - \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)) \\ &= e^{\mu_i + \frac{c_i^2}{2}}(1 - \Phi_{0,1}\left(\frac{\mu_i}{c_i} + c_i\right)) - 1 + \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) \end{aligned} \tag{91}$$

Thus, we obtain:

$$\begin{aligned}\mathbb{E}[Y_i] &= c_i\phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \mu_i\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \lambda e^{\mu_i+\frac{c_i^2}{2}}(1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}+c_i\right)) \\ &- \lambda(1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right))\end{aligned} \tag{92}$$

By noting: $\phi_i = \phi_{0,1}\left(\frac{\mu_i}{c_i}\right)$, $\Phi_i = \Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)$ and $\Phi_{c_i} = \Phi_{0,1}\left(\frac{\mu_i}{c_i}+c_i\right)$ we have:

$$\boxed{\mathbb{E}[Y_i] = c_i\phi_i + (\mu_i+\lambda)\Phi_i + \lambda e^{\mu_i+\frac{c_i^2}{2}}(1-\Phi_{c_i}) - \lambda} \tag{93}$$

5.2. Variance value estimation

$$\begin{aligned}\mathbb{E}[Y_i^2] &= \mathbb{E}_{z\sim p(Z)}\left[f_i(Q_r z+\mu)^2\right] \\ &= \int_{-\infty}^{+\infty} f_i(Q_r z+\mu)^2\phi_{0,\Lambda_r}(z)dz \\ &= \int_{Z_i^+}(z^T q_i+\mu_i)^2\phi_{0,\Lambda_r}(z)dz + \lambda^2\int_{Z_i^-}(e^{z^T q_i+\mu_i}-1)^2\phi_{0,\Lambda_r}(z)dz\end{aligned}$$

Yet, by the equation (23):

$$\int_{Z_i^+}(z^T q_i+\mu_i)^2\phi_{0,\Lambda_r}(z)dz = (\mu_i^2+c_i^2)\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) + \mu_i c_i\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)$$

Now:

$$\begin{aligned}\int_{Z_i^-}(e^{z^T q_i+\mu_i}-1)^2\phi_{0,\Lambda_r}(z)dz &= e^{2\mu_i}\int_{Z_i^-} e^{2z^T q_i}\phi_{0,\Lambda_r}(z)dz \\ &- 2e^{\mu_i}\int_{Z_i^-} e^{z^T q_i}\phi_{0,\Lambda_r}(z)dz + \int_{Z_i^-}\phi_{0,\Lambda_r}(z)dz\end{aligned} \tag{94}$$

with:

$$\int_{Z_i^-}\phi_{0,\Lambda_r}(z)dz = 1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right) \tag{95}$$

$$\begin{aligned}\int_{Z_i^-} e^{z^T q_i}\phi_{0,\Lambda_r}(z)dz &= \int_{-\infty}^{-\frac{\mu_i}{c_i}} e^{c_i b_r}\phi_{0,1}(b_r)db_r \\ &= e^{\frac{c_i^2}{2}}\int_{-\infty}^{-\frac{\mu_i}{c_i}}\phi_{0,1}(b_r-c_i)db_r \\ &= e^{\frac{c_i^2}{2}}(1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}+c_i\right))\end{aligned} \tag{96}$$

$$\begin{aligned}\int_{Z_i^-} e^{2z^T q_i}\phi_{0,\Lambda_r}(z)dz &= e^{2c_i^2}\int_{-\infty}^{-\frac{\mu_i}{c_i}}\phi_{0,1}(b_r-2c_i)db_r \\ &= e^{2c_i^2}(1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}+2c_i\right))\end{aligned} \tag{97}$$

Then:

$$\begin{aligned}\int_{Z_i^-}(e^{z^T q_i+\mu_i}-1)^2\phi_{0,\Lambda_r}(z)dz &= e^{2(c_i^2+\mu_i)}(1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}+2c_i\right)) \\ &- 2e^{\mu_i+\frac{c_i^2}{2}}(1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}+c_i\right))+1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)\end{aligned}$$

By noting: $\phi_i=\phi_{0,1}\left(\frac{\mu_i}{c_i}\right)$, $\Phi_i=\Phi_{0,1}\left(\frac{\mu_i}{c_i}\right)$, $\Phi_{c_i}=\Phi_{0,1}\left(\frac{\mu_i}{c_i}+c_i\right)$ and $\Phi_{2c_i}=\Phi_{0,1}\left(\frac{\mu_i}{c_i}+2c_i\right)$ we have:

$$\begin{aligned}\mathbb{E}[Y_i^2] &= (\mu_i^2+c_i^2)\Phi_i+\mu_i c_i\phi_i+\lambda^2 e^{2(c_i^2+\mu_i)}(1-\Phi_{2c_i})-2\lambda^2 e^{\mu_i+\frac{c_i^2}{2}}(1-\Phi_{c_i}) \\ &+ \lambda^2(1-\Phi_i)\end{aligned} \tag{98}$$

We finally obtain:

$$\boxed{\mathbb{E}[Y_i^2]=(\mu_i^2+c_i^2-\lambda^2)\Phi_i+\mu_i c_i\phi_i+\lambda^2 e^{2(c_i^2+\mu_i)}(1-\Phi_{2c_i})-2\lambda^2 e^{\mu_i+\frac{c_i^2}{2}}(1-\Phi_{c_i})+\lambda^2} \tag{99}$$

5.3. Covariance value estimation

Let's note f_{ij} such that:

$$\begin{aligned}f_{ij}\colon \mathbb{R}^n &\to \mathbb{R}_+ \\ x &\mapsto f_i(x)f_j(x)\end{aligned}$$

with f_i defined as in (89).

$$\begin{aligned}\mathbb{E}[Y_iY_j] &= \mathbb{E}_{z\sim p(Z)}\left[f_{ij}(Q_r z+\mu)\right] \\ &= \mathbb{E}_{z\sim p(Z)}\left[f_i(Q_r z+\mu)f_j(Q_r z+\mu)\right]\end{aligned} \tag{100}$$

By noting:

$$\begin{aligned}Z_{ij}^+ &= \left\{z\in\mathbb{R}^r : z^T q_i+\mu_i\geq 0,\ z^T q_j+\mu_j\geq 0\right\} \\ Z_{ij}^- &= \left\{z\in\mathbb{R}^r : z^T q_i+\mu_i< 0,\ z^T q_j+\mu_j< 0\right\} \\ Z_{ij}^{-+} &= \left\{z\in\mathbb{R}^r : z^T q_i+\mu_i\geq 0,\ z^T q_j+\mu_j< 0\right\} \\ Z_{ij}^{+-} &= \left\{z\in\mathbb{R}^r : z^T q_i+\mu_i< 0,\ z^T q_j+\mu_j\geq 0\right\}\end{aligned}$$

We have:

$$
\begin{aligned}
\mathbb{E}[Y_iY_j] &= \int_{z_{ij}^{+}} f_{ij}(Q_r z+\mu)\phi_{0,\Lambda_r}(z)dz \\
&+ \int_{z_{ij}^{-}} f_{ij}(Q_r z+\mu)\phi_{0,\Lambda_r}(z)dz \\
&+ \int_{z_{ij}^{+-}} f_{ij}(Q_r z+\mu)\phi_{0,\Lambda_r}(z)dz \\
&+ \int_{z_{ij}^{-+}} f_{ij}(Q_r z+\mu)\phi_{0,\Lambda_r}(z)dz \qquad (101)\\
&= (+)+(-)+(+-)+(-+) \qquad (102)
\end{aligned}
$$

5.3.1. DERIVATION OVER Z_{ij}^{-}:

$$
\begin{aligned}
(-) &= \int_{z_{ij}^{-}} f_{ij}(Q_r z+\mu)\phi_{0,\Lambda_r}(z)dz \\
&= \lambda^2 \int_{z_{ij}^{-}} (e^{z^T q_i+\mu_i}-1)(e^{z^T q_j+\mu_j}-1)\phi_{0,\Lambda_r}(z)dz \\
&= \lambda^2[e^{\mu_i+\mu_j}\int_{z_{ij}^{-}} e^{z^T q_i+z^T q_j}\phi_{0,\Lambda_r}(z)dz - e^{\mu_i}\int_{z_{ij}^{-}} e^{z^T q_i}\phi_{0,\Lambda_r}(z)dz \\
&- e^{\mu_j}\int_{z_{ij}^{-}} e^{z^T q_j}\phi_{0,\Lambda_r}(z)dz + \int_{z_{ij}^{-}}\phi_{0,\Lambda_r}(z)dz] \\
&= \lambda^2[e^{\mu_i+\mu_j}(-*) - e^{\mu_i}(-**)_i - e^{\mu_j}(-**)_j + (-***)] \qquad (103)
\end{aligned}
$$

By equation (59), we have:

$$
(-***) = 1-\Phi_i-\Phi_{ij} \qquad (104)
$$

$$
\begin{aligned}
(-**)_i &= \int_{z_{ij}^{-}} e^{z^T q_i}\phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}}\int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} e^{c_i b_r}\phi_{0,1}(b_r)\phi_{0,1}(b_{r-1})db_r db_{r-1} \\
&= e^{\frac{c_i^2}{2}}\int_{-\infty}^{-\frac{\mu_i}{c_i}}\phi_{0,1}(b_r-c_i)\Phi_{0,1}\left(-\frac{\mu_j+\beta b_r}{\alpha}\right)db_r \\
&= e^{\frac{c_i^2}{2}}\left[\int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i}\phi_{0,1}(b_r)db_r - \int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i}\phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j+\beta c_i+\beta b_r}{\alpha}\right)db_r\right] \\
&= e^{\frac{c_i^2}{2}}\left(1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}+c_i\right)-\Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j+\beta c_i}{c_j},-\frac{\mu_i}{c_i}-c_i\right]^T\right)\right) \\
&= e^{\frac{c_i^2}{2}}(1-\Phi_{c_i}-\Phi_A) \qquad (105)
\end{aligned}
$$

where $\Phi_A = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j+\beta c_i}{c_j},-\frac{\mu_i}{c_i}-c_i\right]^T\right)$

$$
\begin{aligned}
(-**)_j &= \int_{z_{ij}^-} e^{z^T q_j} \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} e^{\alpha b_{r-1}+\beta b_r} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_{r-1} db_r \\
&= e^{\frac{\alpha^2+\beta^2}{2}} \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r-\beta) \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} \phi_{0,1}(b_{r-1}-\alpha) db_{r-1} db_r \\
&= e^{\frac{c_j^2}{2}} \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r-\beta)\Phi_{0,1}\left(-\frac{\mu_j+\alpha^2+\beta b_r}{\alpha}\right) db_r \\
&= e^{\frac{c_j^2}{2}} \int_{-\infty}^{-\frac{\mu_i}{c_i}-\beta} \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+c_j^2+\beta b_r}{\alpha}\right) db_r \\
&= e^{\frac{c_j^2}{2}} \left(1-\Phi_{0,1}\left(\frac{\mu_i}{c_i}+\beta\right) - \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j}+c_j, -\frac{\mu_i}{c_i}-\beta\right]^T\right)\right) \\
&= e^{\frac{c_j^2}{2}} \left(1-\Phi_{\beta_i}-\Phi_B\right) \qquad (106)
\end{aligned}
$$

where $\Phi_{\beta_i} = \Phi_{0,1}\left(\frac{\mu_i}{c_i}+\beta\right)$, $\Phi_B = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j}+c_j, -\frac{\mu_i}{c_i}-\beta\right]^T\right)$ and:

$$
\begin{aligned}
(-*) &= \int_{z_{ij}^-} e^{z^T q_i + z^T q_j} \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} e^{c_i b_r+\alpha b_{r-1}+\beta b_r} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_{r-1} db_r \\
&= e^{\frac{\alpha^2+(c_i+\beta)^2}{2}} \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r-(c_i+\beta)) \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} \phi_{0,1}(b_{r-1}-\alpha) db_{r-1} db_r \\
&= e^{\frac{c_j^2+c_i^2+2c_i\beta}{2}} \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r-(c_i+\beta))\Phi_{0,1}\left(-\frac{\mu_j+\alpha^2+\beta b_r}{\alpha}\right) db_r \\
&= e^{\frac{c_j^2+c_i^2}{2}+c_i\beta} \int_{-\infty}^{-\frac{\mu_i}{c_i}-(c_i+\beta)} \phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j+c_j^2+\beta c_i+\beta b_r}{\alpha}\right) db_r \\
&= e^{\frac{c_j^2+c_i^2}{2}+c_i\beta}(1-\Phi_{0,1}(\frac{\mu_i}{c_i}+c_i+\beta) \\
&- \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j+c_j^2+\beta c_i}{c_j}, -(\frac{\mu_i}{c_i}+c_i+\beta)\right]^T\right)) \qquad (107) \\
&= e^{\frac{c_j^2+c_i^2}{2}+c_i\beta}(1-\Phi_{\beta+c_i}-\Phi_C) \qquad (108)
\end{aligned}
$$

where: $\Phi_{\beta+c_i} = \Phi_{0,1}(\frac{\mu_i}{c_i}+c_i+\beta)$ and

$\Phi_C = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j+c_j^2+\beta c_i}{c_j}, -(\frac{\mu_i}{c_i}+c_i+\beta)\right]^T\right)$

Finally:

$$\begin{aligned}(-) &= \lambda^2[e^{\mu_i+\mu_j}(-*) - e^{\mu_i}(-**)_i - e^{\mu_j}(-**)_j + (-***)] \\ &= \lambda^2[e^{\mu_i+\mu_j+\frac{c_j^2+c_i^2}{2}+c_i\beta}(1-\Phi_{\beta+c_i}-\Phi_C) \\ &- e^{\mu_i+\frac{c_i^2}{2}}(1-\Phi_{c_i}-\Phi_A) - e^{\mu_j+\frac{c_j^2}{2}}(1-\Phi_{\beta_i}-\Phi_B) + 1 - \Phi_i - \Phi_{ij}] \end{aligned} \tag{109}$$

where $\Phi_A = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j+\beta c_i}{c_j}, -\frac{\mu_i}{c_i}-c_i\right]^T\right)$,
$\Phi_B = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j}+c_j, -\frac{\mu_i}{c_i}-\beta\right]^T\right)$,
$\Phi_C = \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j+c_j^2+\beta c_i}{c_j}, -(\frac{\mu_i}{c_i}+c_i+\beta)\right]^T\right)$),
and $\Phi_{\beta+c_i} = \Phi_{0,1}(\frac{\mu_i}{c_i}+c_i+\beta)$, $\Phi_{\beta_i} = \Phi_{0,1}\left(\frac{\mu_i}{c_i}+\beta\right)$ and $\Phi_{c_i} = \Phi_{0,1}\left(\frac{\mu_i}{c_i}+c_i\right)$

5.3.2. DERIVATION OVER Z_{ij}^{+-}:

$$\begin{aligned}(+-) &= \int_{z_{ij}^{+-}} f_{ij}(Q_r z+\mu)\phi_{0,\Lambda_r}(z)dz \\ &= \lambda\int_{z_{ij}^{+-}} (z^T q_i+\mu_i)(e^{z^T q_j+\mu_j}-1)\phi_{0,\Lambda_r}(z)dz \\ &= \lambda[e^{\mu_j}\int_{z_{ij}^{+-}} z^T q_i e^{z^T q_j}\phi_{0,\Lambda_r}(z)dz - \int_{z_{ij}^{+-}} z^T q_i\phi_{0,\Lambda_r}(z)dz \\ &+ \mu_i e^{\mu_j}\int_{z_{ij}^{+-}} e^{z^T q_j}\phi_{0,\Lambda_r}(z)dz - \mu_i\int_{z_{ij}^{+-}}\phi_{0,\Lambda_r}(z)dz] \\ &= \lambda[e^{\mu_j}(+-*) - (+-**)_i + \mu_i e^{\mu_j}(+-**)_j - \mu_i(+-***)] \end{aligned} \tag{110}$$

With (76), we have:

$$(+-***) = \Phi_i - \Phi_j + \Phi_{ij} \tag{111}$$

By the equation (74), we have:

$$\begin{aligned}(+-**)_i &= \int_{z_{ij}^{+-}} z^T q_i\phi_{0,\Lambda_r}(z)dz \\ &= c_i\phi_i(1-\Phi_{NS}) - \frac{\beta c_i}{c_j}\phi_j\Phi_S \end{aligned} \tag{112}$$

and:

$$
\begin{aligned}
(+-**)_j &= \int_{z_{ij}^{+-}} e^{z^T q_j} \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} e^{\alpha b_{r-1}+\beta b_r} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_{r-1} db_r \\
&= e^{\frac{c_j^2}{2}} \int_{-\frac{\mu_i}{c_i}}^{+\infty} \phi_{0,1}(b_r-\beta) \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} \phi_{0,1}(b_{r-1}-\alpha) db_{r-1} db_r \\
&= e^{\frac{c_j^2}{2}} \int_{-\frac{\mu_i}{c_i}-\beta}^{+\infty} \phi_{0,1}(b_r) \Phi_{0,1}\left(-\frac{\mu_j+c_j^2+\beta b_r}{\alpha}\right) db_r \\
&= e^{\frac{c_j^2}{2}} (\Phi_{0,1}\left(\frac{\mu_i}{c_i}+\beta\right) - \Phi_{0,1}\left(\frac{\mu_j}{c_j}+c_j\right) \\
&+ \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j}+c_j, -(\frac{\mu_i}{c_i}+\beta)\right]^T\right)) \qquad (113) \\
&= e^{\frac{c_j^2}{2}} (\Phi_{\beta_i} - \Phi_{c_j} + \Phi_B) \qquad (114)
\end{aligned}
$$

and:

$$
\begin{aligned}
(+-*) &= \int_{z_{ij}^{+-}} z^T q_i e^{z^T q_j} \phi_{0,\Lambda_r}(z) dz \\
&= \int_{-\frac{\mu_i}{c_i}}^{+\infty} \int_{-\infty}^{-\frac{\mu_j+\beta b_r}{\alpha}} (c_i b_r) e^{\alpha b_{r-1}+\beta b_r} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1}) db_r db_{r-1} \\
&= c_i e^{\frac{c_j^2}{2}} \int_{-\frac{\mu_i}{c_i}}^{+\infty} b_r \phi_{0,1}(b_r-\beta) \Phi_{0,1}\left(-\frac{\mu_j+\alpha^2+\beta b_r}{\alpha}\right) db_r \\
&= c_i e^{\frac{c_j^2}{2}} [\int_{-\frac{\mu_i}{c_i}-\beta}^{+\infty} b_r \phi_{0,1}(b_r) \Phi_{0,1}\left(-\frac{\mu_j+c_j^2+\beta b_r}{\alpha}\right) db_r \\
&+ \beta \int_{-\frac{\mu_i}{c_i}-\beta}^{+\infty} \phi_{0,1}(b_r) \Phi_{0,1}\left(-\frac{\mu_j+c_j^2+\beta b_r}{\alpha}\right) db_r] \\
&= c_i e^{\frac{c_j^2}{2}} [(+-*)_1 + \beta(+-*)_2] \qquad (115)
\end{aligned}
$$

With:

$$
\begin{aligned}
(+-*)_2 &= \int_{-\frac{\mu_i}{c_i}-\beta}^{+\infty} \phi_{0,1}(b_r)db_r - \int_{-\frac{\mu_i}{c_i}-\beta}^{+\infty} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j + c_j^2 + \beta b_r}{\alpha}\right) db_r \\
&= \int_{-\frac{\mu_i}{c_i}-\beta}^{+\infty} \phi_{0,1}(b_r)db_r - \int_{-\infty}^{+\infty} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j + c_j^2 + \beta b_r}{\alpha}\right) db_r \\
&+ \int_{-\infty}^{-\frac{\mu_i}{c_i}-\beta} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j + c_j^2 + \beta b_r}{\alpha}\right) db_r \\
&= \Phi_{0,1}\left(\frac{\mu_i}{c_i} + \beta\right) - \Phi_{0,1}\left(\frac{\mu_j}{c_j} + c_j\right) \\
&+ \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j}{c_j} + c_j, -(\frac{\mu_i}{c_i} + \beta)\right]^T\right) \\
&= \Phi_{\beta_i} - \Phi_{c_j} + \Phi_B
\end{aligned} \tag{116}
$$

and:

$$
\begin{aligned}
(+-*)_1 &= \left[-\phi_{0,1}(b_r)\Phi_{0,1}\left(-\frac{\mu_j + c_j^2 + \beta b_r}{\alpha}\right)\right]_{-\frac{\mu_i}{c_i}-\beta}^{+\infty} \\
&- \frac{\beta}{\alpha}\int_{-\frac{\mu_i}{c_i}-\beta}^{+\infty} \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j + c_j^2 + \beta b_r}{\alpha}\right) db_r \\
&= \phi_{0,1}\left(\frac{\mu_i}{c_i} + \beta\right)\Phi_{0,1}\left(-\frac{\mu_j + c_j^2 - \beta^2}{\alpha} + \frac{\beta\mu_i}{\alpha c_i}\right) \\
&- \frac{\beta}{c_j}\phi_{0,1}\left(\frac{\mu_j}{c_j} + c_j\right)\Phi_{0,1}\left(\frac{c_j\mu_i}{\alpha c_i} + \frac{\beta c_j}{\alpha} - \frac{\beta}{\alpha}(\frac{\mu_j}{c_j} + c_j)\right) \\
&= \phi_{\beta_i}\Phi_{0,1}\left(\frac{\beta\mu_i}{\alpha c_i} - \frac{\mu_j}{\alpha} - \alpha\right) - \frac{\beta}{c_j}\Phi_{0,1}\left(\frac{c_j\mu_i}{\alpha c_i} - \frac{\beta\mu_j}{\alpha c_j}\right) \\
&= \phi_{\beta_i}\Phi_\alpha - \frac{\beta}{c_j}\phi_{c_j}\Phi_S
\end{aligned} \tag{117}
$$

With $\Phi_\alpha = \Phi_{0,1}\left(\frac{\beta\mu_i}{\alpha c_i} - \frac{\mu_j}{\alpha} - \alpha\right)$.

Then:

$$
(+-*) = c_i e^{\frac{c_j^2}{2}}[\phi_{\beta_i}\Phi_\alpha - \frac{\beta}{c_j}\phi_{c_j}\Phi_S + \beta(\Phi_{\beta_i} - \Phi_{c_j} + \Phi_B)] \tag{118}
$$

Thus, we have:

$$
\begin{aligned}
(+-) &= \lambda[e^{\mu_j}(+-*) - (+-**)_i + \mu_i e^{\mu_j}(+-**)_j - \mu_i(+-***)] \\
&= \lambda[c_i e^{\mu_j + \frac{c_j^2}{2}}(\phi_{\beta_i}\Phi_\alpha - \frac{\beta}{c_j}\phi_{c_j}\Phi_S + \beta(\Phi_{\beta_i} - \Phi_{c_j} + \Phi_B)) - c_i\phi_i(1 - \Phi_{NS}) \\
&+ \frac{\beta c_i}{c_j}\phi_j\Phi_S + \mu_i e^{\mu_j + \frac{c_j^2}{2}}(\Phi_{\beta_i} - \Phi_{c_j} + \Phi_B) - \mu_i(\Phi_i - \Phi_j + \Phi_{ij})] \\
&= \lambda[e^{\mu_j + \frac{c_j^2}{2}}(\mu_i + \beta c_i)(\Phi_{\beta_i} - \Phi_{c_j} + \Phi_B) + \frac{\beta c_i}{c_j}(\phi_j - e^{\mu_j + \frac{c_j^2}{2}}\phi_{c_j})\Phi_S \\
&+ c_i e^{\mu_j + \frac{c_j^2}{2}}\phi_{\beta_i}\Phi_\alpha - c_i\phi_i(1 - \Phi_{NS}) - \mu_i(\Phi_i - \Phi_j + \Phi_{ij})]
\end{aligned} \tag{119}
$$

5.3.3. DERIVATION OVER Z_{ij}^{-+}:

$$\begin{aligned}
(-+) &= \int_{z_{ij}^{-+}} f_{ij}(Q_r z + \mu)\phi_{0,\Lambda_r}(z)dz \\
&= \lambda \int_{z_{ij}^{-+}} (e^{z^T q_i + \mu_i} - 1)(z^T q_j + \mu_j)\phi_{0,\Lambda_r}(z)dz \\
&= \lambda[e^{\mu_i} \int_{z_{ij}^{-+}} e^{z^T q_i} z^T q_j \phi_{0,\Lambda_r}(z)dz - \int_{z_{ij}^{-+}} z^T q_j \phi_{0,\Lambda_r}(z)dz \\
&+ \mu_j e^{\mu_i} \int_{z_{ij}^{-+}} e^{z^T q_i} \phi_{0,\Lambda_r}(z)dz - \mu_j \int_{z_{ij}^{-+}} \phi_{0,\Lambda_r}(z)dz]
\end{aligned} \tag{120}$$

With (85), we have:

$$(-+***) = \Phi_{ij} \tag{121}$$

By the equation (84), we have:

$$\begin{aligned}
(-+**)_j &= \int_{z_{ij}^{-+}} z^T q_j \phi_{0,\Lambda_r}(z)dz \\
&= -\beta\phi_i\Phi_{NS} + c_j\phi_j(1 - \Phi_S)
\end{aligned} \tag{122}$$

and:

$$\begin{aligned}
(-+**)_i &= \int_{z_{ij}^{-+}} e^{z^T q_i} \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\frac{\mu_j+\beta b_r}{\alpha}}^{+\infty} e^{c_i b_r} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1})db_{r-1}db_r \\
&= e^{\frac{c_i^2}{2}} \int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j + \beta c_i + \beta b_r}{\alpha}\right) db_r \\
&= e^{\frac{c_i^2}{2}} \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j+\beta c_i}{c_j}, -(\frac{\mu_i}{c_i} + c_i)\right]^T\right) \\
&= e^{\frac{c_i^2}{2}} \Phi_A
\end{aligned} \tag{123}$$

Finally,

$$\begin{aligned}
(-+*) &= \int_{z_{ij}^{-+}} z^T q_j e^{z^T q_i} \phi_{0,\Lambda_r}(z)dz \\
&= \int_{-\infty}^{-\frac{\mu_i}{c_i}} \int_{-\frac{\mu_j+\beta b_r}{\alpha}}^{+\infty} (\alpha b_{r-1} + \beta b_r) e^{c_i b_r} \phi_{0,1}(b_r)\phi_{0,1}(b_{r-1})db_{r-1}db_r \\
&= e^{\frac{c_i^2}{2}}[\alpha \int_{-\infty}^{-\frac{\mu_i}{c_i}} \phi_{0,1}(b_r - c_i)\phi_{0,1}\left(\frac{\mu_j + \beta b_r}{\alpha}\right) db_r \\
&+ \beta \int_{-\infty}^{-\frac{\mu_i}{c_i}} b_r \phi_{0,1}(b_r - c_i)\Phi_{0,1}\left(\frac{\mu_j + \beta b_r}{\alpha}\right) db_r]
\end{aligned} \tag{124}$$

With:

$$\begin{aligned}
\int_{-\infty}^{-\frac{\mu_i}{c_i}} & \phi_{0,1}(b_r - c_i)\phi_{0,1}\left(\frac{\mu_j + \beta b_r}{\alpha}\right) db_r = \int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i} \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j + \beta c_i + \beta b_r}{\alpha}\right) db_r \\
= & \phi_{0,1}\left(\frac{\mu_j + \beta c_i}{c_j}\right) \int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i} \phi_{0,1}\left(\frac{c_j}{\alpha} b_r + \frac{\beta}{\alpha}\frac{\mu_j + \beta c_i}{c_j}\right) db_r \\
= & \frac{\alpha}{c_j}\phi_{0,1}\left(\frac{\mu_j + \beta c_i}{c_j}\right)\Phi_{0,1}\left(\frac{\beta}{\alpha}\frac{\mu_j + \beta c_i}{c_j} - \frac{c_j\mu_i}{\alpha c_i} - \frac{c_j c_i}{\alpha}\right)
\end{aligned} \tag{125}$$

and:

$$\begin{aligned}
\int_{-\infty}^{-\frac{\mu_i}{c_i}} & b_r\phi_{0,1}(b_r - c_i)\Phi_{0,1}\left(\frac{\mu_j + \beta b_r}{\alpha}\right) db_r = \int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i} b_r\phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j + \beta c_i + \beta b_r}{\alpha}\right) db_r \\
+ & c_i \int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j + \beta c_i + \beta b_r}{\alpha}\right) db_r
\end{aligned}$$

With:

$$\int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i} \phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j + \beta c_i + \beta b_r}{\alpha}\right) db_r = \Phi_A \tag{126}$$

and:

$$\begin{aligned}
\int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i} & b_r\phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j + \beta c_i + \beta b_r}{\alpha}\right) db_r = \left[-\phi_{0,1}(b_r)\Phi_{0,1}\left(\frac{\mu_j + \beta c_i + \beta b_r}{\alpha}\right)\right]_{-\infty}^{-\frac{\mu_i}{c_i}-c_i} \\
+ & \frac{\beta}{\alpha}\int_{-\infty}^{-\frac{\mu_i}{c_i}-c_i} \phi_{0,1}(b_r)\phi_{0,1}\left(\frac{\mu_j + \beta c_i + \beta b_r}{\alpha}\right) db_r \\
= & -\phi_{0,1}\left(\frac{\mu_i}{c_i} + c_i\right)\Phi_{0,1}\left(\frac{\mu_j}{\alpha} - \frac{\beta\mu_i}{\alpha c_i}\right) \\
+ & \frac{\beta}{c_j}\phi_{0,1}\left(\frac{\mu_j + \beta c_i}{c_j}\right)\Phi_{0,1}\left(\frac{\beta}{\alpha}\frac{\mu_j + \beta c_i}{c_j} - \frac{c_j\mu_i}{\alpha c_i} - \frac{c_j c_i}{\alpha}\right)
\end{aligned}$$

Then:

$$\begin{aligned}
(-+*) = & e^{\frac{c_i^2}{2}}[\frac{\alpha^2 + \beta^2}{c_j}\phi_{0,1}\left(\frac{\mu_j + \beta c_i}{c_j}\right)\Phi_{0,1}\left(\frac{\beta}{\alpha}\frac{\mu_j + \beta c_i}{c_j} - \frac{c_j\mu_i}{\alpha c_i} - \frac{c_j c_i}{\alpha}\right) \\
- & \beta\phi_{0,1}\left(\frac{\mu_i}{c_i} + c_i\right)\Phi_{0,1}\left(\frac{\mu_j}{\alpha} - \frac{\beta\mu_i}{\alpha c_i}\right) \\
+ & \beta c_i \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j + \beta c_i}{c_j}, -(\frac{\mu_i}{c_i} + c_i)\right]^T\right)] \\
= & e^{\frac{c_i^2}{2}}[c_j\phi_{0,1}\left(\frac{\mu_j + \beta c_i}{c_j}\right)\Phi_{0,1}\left(\frac{\beta}{\alpha}\frac{\mu_j + \beta c_i}{c_j} - \frac{c_j\mu_i}{\alpha c_i} - \frac{c_j c_i}{\alpha}\right) \\
- & \beta\phi_{0,1}\left(\frac{\mu_i}{c_i} + c_i\right)\Phi_{0,1}\left(\frac{\mu_j}{\alpha} - \frac{\beta\mu_i}{\alpha c_i}\right) \\
+ & \beta c_i \Phi_{\begin{bmatrix}0\\0\end{bmatrix},\begin{bmatrix}1 & -\frac{\beta}{c_j}\\ -\frac{\beta}{c_j} & 1\end{bmatrix}}\left(\left[\frac{\mu_j + \beta c_i}{c_j}, -(\frac{\mu_i}{c_i} + c_i)\right]^T\right)] \\
= & e^{\frac{c_i^2}{2}}[c_j\phi_a\Phi_e - \beta\phi_{c_i}\Phi_{NS} + \beta c_i\Phi_A]
\end{aligned} \tag{127}$$

where $\phi_a = \phi_{0,1}\left(\frac{\mu_j+\beta c_i}{c_j}\right)$ and $\Phi_e = \Phi_{0,1}\left(\frac{\beta}{\alpha}\frac{\mu_j+\beta c_i}{c_j} - \frac{c_j\mu_i}{\alpha c_i} - \frac{c_j c_i}{\alpha}\right)$.

Then:

$$\begin{aligned}
(-+) &= \lambda[e^{\mu_i}(-+*) - (-+**)_j + \mu_j e^{\mu_i}(-+**)_i - \mu_j(-+***)] \\
&= \lambda[e^{\mu_i+\frac{c_i^2}{2}}(c_j\phi_a\Phi_e - \beta\phi_{c_i}\Phi_{NS} + \beta c_i\Phi_A) + \beta\phi_i\Phi_{NS} - c_j\phi_j(1-\Phi_S) \\
&+ \mu_j e^{\mu_i+\frac{c_i^2}{2}}\Phi_A - \mu_j\Phi_{ij}] \\
&= \lambda[e^{\mu_i+\frac{c_i^2}{2}}(\beta c_i + \mu_j)\Phi_A + \beta(\phi_i - e^{\mu_i+\frac{c_i^2}{2}}\phi_{c_i})\Phi_{NS} + c_j e^{\mu_i+\frac{c_i^2}{2}}\phi_a\Phi_e \\
&- c_j\phi_j(1-\Phi_S) - \mu_j\Phi_{ij}]
\end{aligned} \tag{128}$$

5.3.4. FINAL DERIVATION:

Finally, by the equation (46), we have:

$$(+) = \mathbb{E}[Y_iY_j] = c_i\alpha\phi_j\phi_{NS} + (\mu_i\mu_j + c_i\beta)(\Phi_j - \Phi_{ij}) + \mu_i c_j\phi_j\Phi_S + \mu_j c_i\phi_i\Phi_{NS}$$

Then:

$$\mathbb{E}[Y_iY_j] = (+) + (-) + (+-) + (-+) \tag{129}$$

Finally, we obtain:

$$\begin{aligned}
\mathbb{E}[Y_iY_j] &= c_i\alpha\phi_j\phi_{NS} + (\mu_i\mu_j + c_i\beta)(\Phi_j - \Phi_{ij}) + \mu_i c_j\phi_j\Phi_S + \mu_j c_i\phi_i\Phi_{NS} \\
&+ \lambda^2[e^{\mu_i+\mu_j+\frac{c_j^2+c_i^2}{2}+c_i\beta}(1 - \Phi_{\beta+c_i} - \Phi_C) \\
&- e^{\mu_i+\frac{c_i^2}{2}}(1-\Phi_{c_i}-\Phi_A) - e^{\mu_j+\frac{c_j^2}{2}}(1-\Phi_{\beta_i}-\Phi_B) + 1 - \Phi_i - \Phi_{ij}] \\
&+ \lambda[e^{\mu_j+\frac{c_j^2}{2}}(\mu_i+\beta c_i)(\Phi_{\beta_i} - \Phi_{c_j} + \Phi_B) + \frac{\beta c_i}{c_j}(\phi_j - e^{\mu_j+\frac{c_j^2}{2}}\phi_{c_j})\Phi_S \\
&+ c_i e^{\mu_j+\frac{c_j^2}{2}}\phi_{\beta_i}\Phi_\alpha - c_i\phi_i(1-\Phi_{NS}) - \mu_i(\Phi_i - \Phi_j + \Phi_{ij})] \\
&+ \lambda[e^{\mu_i+\frac{c_i^2}{2}}(\beta c_i + \mu_j)\Phi_A + \beta(\phi_i - e^{\mu_i+\frac{c_i^2}{2}}\phi_{c_i})\Phi_{NS} + c_j e^{\mu_i+\frac{c_i^2}{2}}\phi_a\Phi_e \\
&- c_j\phi_j(1-\Phi_S) - \mu_j\Phi_{ij}]
\end{aligned} \tag{130}$$

Thus:

$$
\begin{aligned}
\mathbb{E}[Y_iY_j] &= c_i\alpha\phi_j\phi_{NS} \\
&+ \left(\mu_i\mu_j + c_i\beta + \lambda\mu_i\right)\Phi_j \\
&- \left(\mu_i\mu_j + c_i\beta + \lambda(\mu_i+\mu_j) + \lambda^2\right)\Phi_{ij} \\
&+ \left(\mu_i c_j\phi_j + \lambda c_j\phi_j + \frac{\lambda\beta c_i}{c_j}((\phi_j - e^{\mu_j+\frac{c_j^2}{2}}\phi_{c_j}))\Phi_S\right. \\
&+ \left(\mu_j c_i\phi_i + \lambda c_i\phi_i + \lambda\beta(\phi_i - e^{\mu_i+\frac{c_i^2}{2}}\phi_{c_i}))\Phi_{NS}\right. \\
&+ \lambda c_i e^{\mu_j+\frac{c_j^2}{2}}\phi_{\beta_i}\Phi_\alpha \\
&+ \lambda c_j e^{\mu_i+\frac{c_i^2}{2}}\phi_a\Phi_e \\
&+ \lambda^2 e^{\mu_i+\frac{c_i^2}{2}}\Phi_{c_i} \\
&- \lambda e^{\mu_j+\frac{c_j^2}{2}}(\mu_i+\beta c_i)\Phi_{c_j} \\
&+ \lambda e^{\mu_i+\frac{c_i^2}{2}}(\lambda+\beta c_i+\mu_j)\Phi_A \\
&+ \lambda e^{\mu_j+\frac{c_j^2}{2}}(\lambda+\beta c_i+\mu_i)(\Phi_{\beta_i}+\Phi_B) \\
&+ \lambda^2 e^{\mu_i+\mu_j+\frac{c_i^2+c_j^2}{2}+c_i\beta}(1-\Phi_{\beta+c_i}-\Phi_C) \\
&- \lambda(\lambda e^{\mu_i+\frac{c_i^2}{2}} + \lambda e^{\mu_j+\frac{c_j^2}{2}} + c_i\phi_i + c_j\phi_j - \lambda)
\end{aligned} \tag{131}
$$

REFERENCES

Clevert, D.-A., Unterthiner, T., and Hochreiter, S. (2015). Fast and accurate deep network learning by exponential linear units (elus). *arXiv preprint arXiv:1511.07289*.

Fukushima, K. (1975). Cognitron: A self-organizing multilayered neural network. *Biological cybernetics*, 20(3):121–136.

Maas, A. L., Hannun, A. Y., Ng, A. Y., et al. (2013). Rectifier nonlinearities improve neural network acoustic models. In *Proc. ICML*, volume 30, page 3.

Owen, D. B. (1980). A table of normal integrals: A table. *Communications in Statistics-Simulation and Computation*, 9(4):389–419.